

F24-190-D-WattWiseAI

Project Team

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Chapter 1

Introduction

Propose here is called WattWiseAI. It is an intelligent system for proper daily electricity demand forecasting in this ever energy-dependent world. A hybrid model, WattWiseAI offers a high potential accuracy based on historical electricity consumption data along with additional season and weather variables. This initiative meets the emerging requirement of high accuracy needed in demand forecasting in today's fast-changing energy landscape for better optimization of supply, less wastage, and sustainability in energy management. This project has many stakeholders in them, starting from the development team that is working on implementing the system, to the FYP Panel that is assessing its progress of work, and to the supervising faculty guiding and evaluating its outcomes. By combining the latest machine learning techniques with real-world data, WattWiseAI aims to extract actionable insights into how electricity demand management strategies are conceptualized and executed worldwide in a meaningful way.

1.1 Problem Statement

It is critically important for error-free prediction of daily demand for electricity and a challenge for the efficient operation and management of power grids. Traditional prediction models, such as statistical methods and standalone machine learning algorithms, more often than not can not capture the complex, nonlinear, and nonstationary patterns present in Electricity demand data. Such limitations cause an error in the forecast which can cause major operational problems such as grid instability, energy waste, and higher operational costs.

Incorrect demand forecasts can result in either underestimation or Overestimating the amount of electricity needed, which then leads to Power supply disruptions or cause to generate unneeded energy. both of which are costly and inefficient. Moreover, there have been studies showing that a 1% increase in electricity load forecast error It can result in

millions of dollars lost.

There should be a more subtle approach in solving these challenges. Predictive models which can much better capture the complexities of electricity demand.

1.2 Scope

The development of WattWiseAI reflects the methodology for developing a hybrid deep-learning model to increase the accuracy of the forecast of electricity demand. This project will integrate ANN with Encoder-Decoder Long Short-Term Memory and ICEEMDAN. These main elements are designed to capture nonlinear patterns, handle sequential data, and filter high-frequency noise from electricity demand data.

The scope entails data collection, model development, training, testing, and even deployment. The team will gather all available historical data from the utility companies and from any public datasets, then pre-process it for handling missing values and normalization before preparing to train a hybrid model. The developed model will be trained and further validated on the data for improving its accuracy and, of course, comparing the results with the already existing solutions from AutoGrid and Grid4C.

Further testing shall be done under multiple scenarios to check on scalability and robustness. Deployment shall cover the development of the user interface that will enable real-time prediction of electricity demand with scenario analysis. Finally, the final model should provide better accuracy in the forecasts, much stability of the grid, and potential cost savings by utility operators. The documentation and reporting will outline the technical details, results, and possible economic benefits for the project.

1.3 Modules

Our system would be divided into the following modules:

1.3.1 Data Collection and Integration

1. This module is responsible for collecting and integrating data from different resources from different companies, government agencies and various public data sets. It is also capable of generating data based on typical patterns and trends if real data is not available or inaccessible.

1.3.2 Data Processing and Analysis

1. This module processes the collected data in various ways to ensure it is ready for training the models. In this module we perform cleaning of data by dealing with missing values and outliers. Furthermore, We will perform resampling and aggregation of data as required for effective model training.

1.3.3 Model Training Module

1. This module is to train a hybrid deep learning model consisting of the components: ANN, EDLSTM and ICEEMDAN. The training process involves feeding the pre-processed data into each model, adjusting model parameters through techniques like backpropagation, and hyperparameter tuning to improve performance.

1.3.4 Prediction Module

1. It is the module that creates electricity demand forecasts using this proposed hybrid deep learning model. It would combine forecast values of the modules of ANN, EDLSTM, and ICEEMDAN to bring about an accurate and reliable forecast. The module would have features that would also reveal the accuracy of the prediction, update the model with new data, and tune up the parameters of the model to ensure long-term performance. Another key feature that enables scenario analysis is such that a forecast can be prepared even under variant conditions like seasonal variations or spikes.

1.3.5 Model Deployment Module

1. This module is actually the deployment of the trained model into either the real world or a simulated environment. This will involve the creation of infrastructural arrangement that would allow the model in a production situation where the efficiency of the model is established within the setting of a real-time situation.

1.3.6 User Interface And Visualization Module

1. This module provides a user-friendly interface to interact with the model and the predictions. It includes visualization tools for the forecast results, time-series plots, error distributions, and scenario analysis. The interface enables users to insert new data, run predictions, and show the results in an accessible format. Another possi-

bility could be offering options for customization, depending on different types of user needs: changing the forecast horizon or choosing specific time periods.

1.4 User Classes and Characteristics

The primary users of the system will include the following classes:

User class	Description
Users	The major end users of the WattWiseAI system will be Electricity Grid Operators, Energy Managers, and Utility Company Analysts. The above-mentioned listed users would utilize the system in observing, recording electricity demand for maintaining stability within the grid, and to assist decision-making processes regarding the distribution of energy. Hence, these users can forecast demands in electricity; they can look into historical data and provide reports facilitated by real-time predictions. The user can input parameters, like date and region, for which the model is going to predict the demanded value.
Administrators	The project of WattWiseAI has an Admin who oversees the entire management and maintenance of the forecasting system. Such Admin duties consist of overseeing data operations, model training, and ensuring optimal performance from the system. They collect and upload historical consumption data and related datasets for model training in the system. They process data cleaning and normalization activities to make raw data available for analysis and training purposes. Admin initiates the training of the hybrid prediction model, training it to learn well from the provided data.

Table 1.1: User Classes

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project. The requirement gathering technique we chose is Use case.

2.1 Use-case

This section includes the Use Case diagram and high level use cases of the entire project along with the expanded use cases for this iteration.

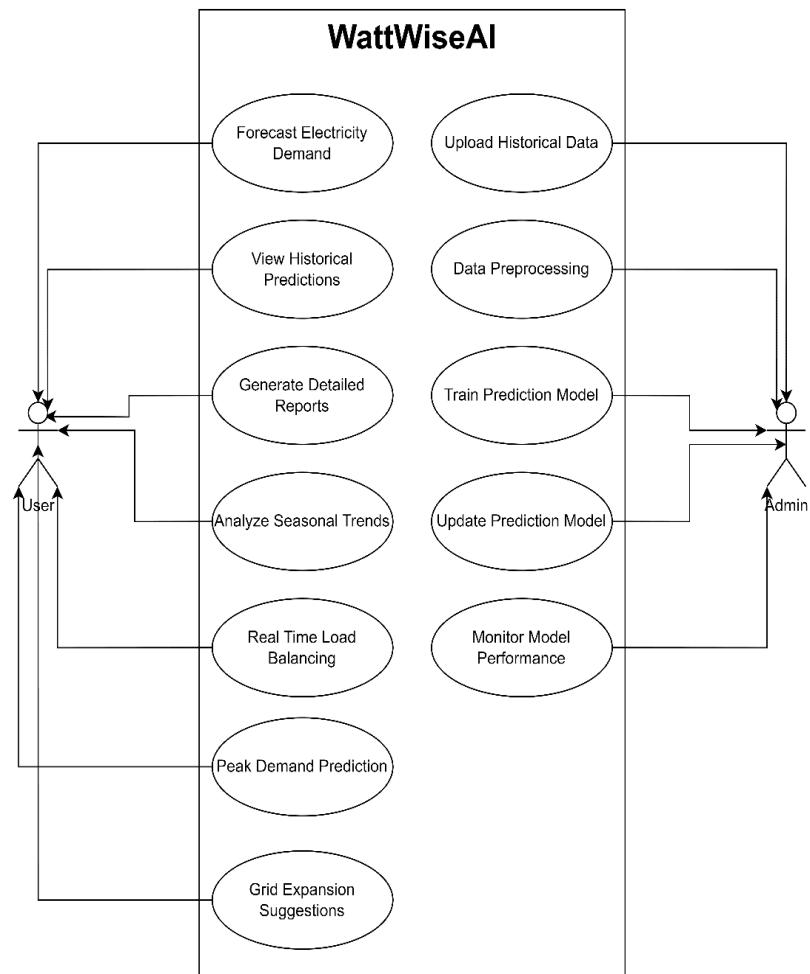


Figure 2.1: Use Case Diagram

UC-01: Forecasting Electricity Demand

Use Case	Forecasting Electricity Demand
Actors	Operators
Type	Primary
Description	Users can input specific parameters (e.g., date, region) to receive predictions for future electricity demand.

UC-02: Historical Predictions

Use Case	View Historical Predictions
Actors	Operators
Type	Primary
Description	Users can access past electricity demand forecasts and compare them with actual consumption data to assess model accuracy.

UC-03: Generate Detailed Reports

Use Case	Generate Detailed Reports
Actors	Operators
Type	Primary
Description	Users can request comprehensive reports summarizing predictions, performance metrics, and historical trends.

UC-04: Analyze Seasonal Trends

Use Case	Analyze Seasonal Trends
Actors	Operators
Type	Secondary
Description	Users can analyze how electricity demand varies by season to identify patterns and inform planning.

UC-05: Real-Time Load Balancing

Use Case	Real-Time Load Balancing
Actors	Operators
Type	Primary

UC-05: Real-Time Load Balancing

Description	The system provides users with recommendations for load balancing based on real-time electricity consumption data.
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UC-06: Peak Demand Prediction

Use Case	Peak Demand Prediction
Actors	Operators
Type	Secondary
Description	Users can forecast peak demand periods, allowing them to prepare for high consumption times.

UC-07: Upload Historical Data

Use Case	Upload Historical Data
Actors	Admin
Type	Primary
Description	Admins upload historical electricity consumption and relevant datasets for model training.

UC-08: Data Preprocessing

Use Case	Data Preprocessing
Actors	Operator
Type	Secondary
Description	Admins pre-process the uploaded data to clean and normalize it, ensuring it is ready for training the model.

UC-09: Train Prediction Model

Use Case	Train Prediction Model
Actors	Operator
Type	Primary
Description	Admins perform the training process for the hybrid model using the pre-processed data.

UC-10: Update Prediction Model	
Use Case	Update Prediction Model
Actors	Operator
Type	Secondary
Description	Admins update the prediction model with new data and improved algorithms to enhance accuracy.
UC-11: Monitor Model Performance	
Use Case	Monitor Model Performance
Actors	Operator
Type	Secondary
Description	Admins monitor the performance of the model, check accuracy metrics and make necessary adjustments.

Table 2.11: Usecases

2.2 Functional Requirements

2.2.1 Data Collection and Integration

Following are the requirements for module 1:

1. The system should be able to gather data from different sources like government sites, public datasets and utility companies
2. The system should also be able to generate synthetic data to improve accuracy

2.2.2 Data Pre-processing

Following are the requirements for module 2:

1. The system should clean the data and handle outliers
2. The system should fill in any missing values

2.2.3 Hybrid Model Implementation

Following are the requirements for module 3:

1. The system shall implement a hybrid deep-learning model that combines ANN, EDLSTM, and ICEEMDAN for predicting electricity demand.
2. The system shall accurately decompose electricity demand data into high- and low-frequency components using ICEEMDAN.

2.2.4 Model Training and Validation

Following are the requirements for module 4:

1. The system shall train the hybrid model using the processed data and validate its performance using cross-validation techniques.
2. The system shall optimize the model's hyperparameters to achieve the best prediction accuracy.

2.2.5 Prediction Module

Following are the requirements for module 5:

1. The system shall provide daily electricity demand predictions based on the trained model.
2. The system shall visualize the prediction results in a user-friendly format via the web interface.

2.2.6 Web Application Interface

Following are the requirements for module 6:

1. The system shall present prediction results and data visualizations through an interactive web interface built with React.js.
2. The system shall allow users to query past predictions and compare them with actual demand data.

2.2.7 Data Storage and Management

Following are the requirements for module 7:

1. The system shall store historical data, model outputs, and user queries in a SQL database.
2. The system shall ensure secure and efficient retrieval of data for analysis and comparison.

2.2.8 Version Control and Collaboration

Following are the requirements for module 8:

1. The system shall use GitHub for version control, enabling collaborative development and tracking changes in the codebase.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements.

2.3.1 Performance

- The system should provide electricity demand predictions with a response time of less than 2 seconds after data input.
- The model training process should complete within a reasonable time frame, depending on the dataset size, with the goal of processing a typical daily dataset within 30 minutes.

2.3.2 Scalability

- The system should be able to handle increasing amounts of data without significant degradation in performance, accommodating future expansion to larger datasets or additional data sources.
- The system should be able to serve multiple users at the same time, ensuring a smooth experience

2.3.3 Reliability

- The system should be up 98% of the time so that users can access the system according to their needs

- The system should be able to recover from crashes efficiently and provide logs to help prevent crashes in future

2.3.4 Usability

- The interface should be easy to use. Operations should follow simple steps, be self explanatory and guide users
- The system should display appropriate messages to guide users in case they make a mistake and need help

2.3.5 Security

- The system should ensure data security by using various protocols and passwords to verify that data is given only to authorized users
- All communication should be secure and encrypted using various techniques to further protect data

2.4 Domain Model

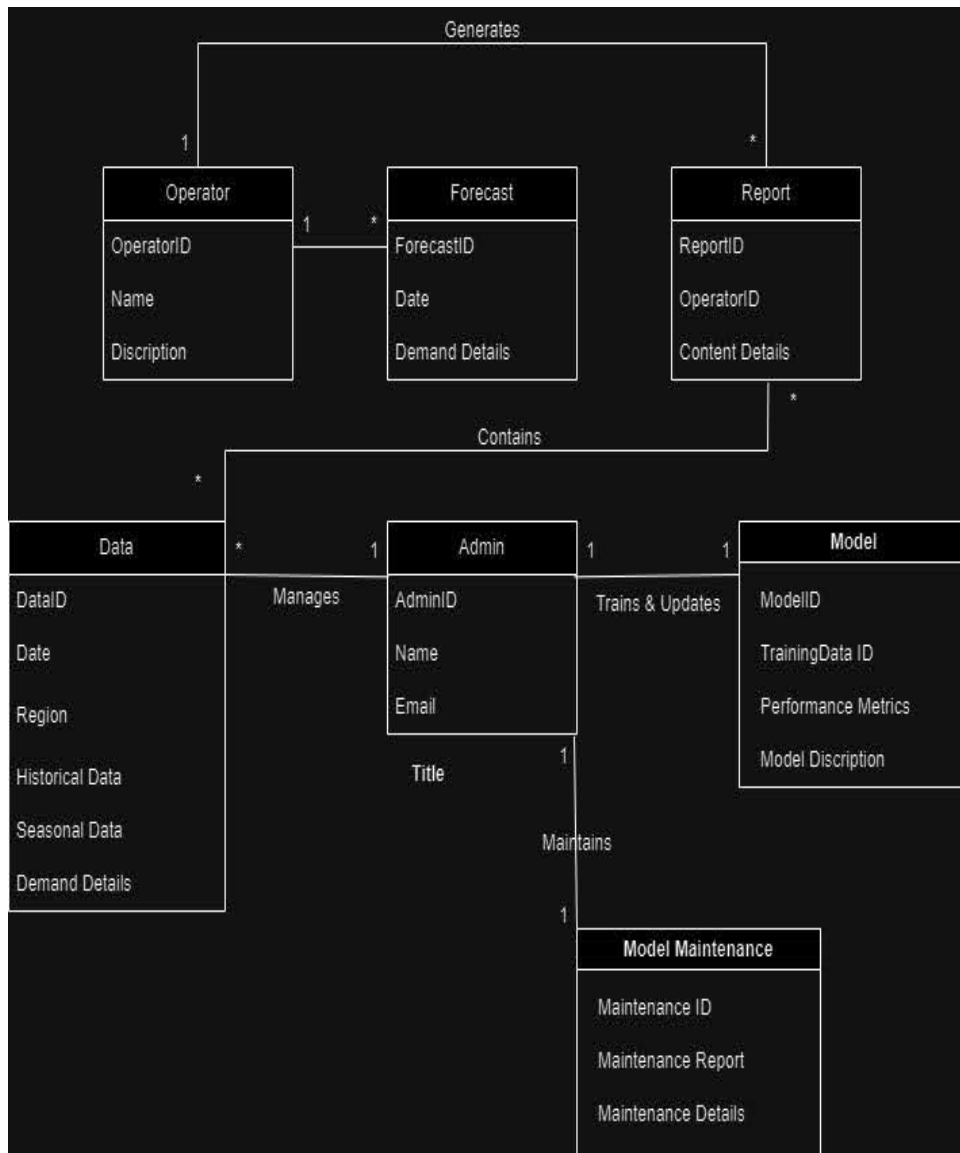


Figure 2.2: Domain Model

Chapter 3

System Overview

The proposed WattWiseAI system will provide accurate, short-term electricity demand forecasting using a state-of-the-art hybrid deep learning model. In general, there are two primary types of users: end-users and administrative users, who manage data and update the prediction models.

At the heart of WattWiseAI, there is in-depth architecture oriented so that every component works in harmony to empower impeccable predictions for electricity demand.

The core engine fuses ANN, EDLSTM, and a technique known as ICEEMDAN. Such a strong hybrid model has been devised for this complex, continuously changing nature of electricity demand data. It provides very high accuracy by decomposing the data into different parts and then using deep learning techniques.

WattWiseAI gives real-time demand prediction with past predictions it generates reports based on seasonal trends. Some advanced functions entail forecasting peak demand and load balancing analytics. Thus, these are some of the key features where one would find useful insights required to make prudent decisions toward grid stability and efficiency.

The administrators manage the system's backend, uploading historical electricity consumption data, preparing and analyzing it, training new prediction models, updating the models based on their performance evaluations, and so on. There are even advanced tools integrated into the system, such as PeakDemandAnalyzer and SeasonalTrendAnalyzer, which give detailed insight into the high-demand periods and seasonal trends, respectively, to help the user draw up reports and alerts on disruption possibilities for fine-tuning their operations.

Furthermore, WattWiseAI combines historical and real-time weather information to enhance the accuracy of its forecasts by correlating such weather conditions with energy consumption to best reflect how environmental conditions impact demand. The system is built in such a way that its architecture is modular and scalable, which makes the maintenance and further enhancement quite easy. The system handles everything concerning

user management, prediction modeling, data analysis, and alert mechanisms separately in their relevant modules. Based on this layered architecture, the system processes oodles of data by running advanced analysis and providing actionable insights to both end-users and administrators.

3.1 Architectural Design

3.1.1 Modular Program Structure

The system WattWiseAI is based on modular architecture. The modules will interact with each other to ensure that efficient data flow and to perform the accurate predictions. This architecture is also very useful for future integration of additional features.

3.1.1.1 Data Management

Functionality: The module provides the facilities of collection of data, storage of data, and preprocessing of historical electricity consumption data and weather related data. This module also supports uploading data files and connects to external sources for real-time updates and suggestions.

Relationship: It works closely with the Model Training and Prediction Modules by supplying clean, prepared and processed data, ensuring that all data used in modeling and forecasting is consistent, clean and processed.

3.1.1.2 Model Training and Updating

Functionality: The module manages the training of the hybrid model, which includes different modules including ANN, EDLSTM, and ICEEMDAN. It includes the training of all the model on the preprocessed data. It also responsible for the updating of the model to increase the accuracy in electricity consumption predictions. Then to integrate the all the models to show the predictions.

Relationship: It Integrates with the Data Management Module to access preprocessed data and sends updated models to the Prediction Module for real-time demand forecasting. This module also monitors the model's performance to ensure its accuracy meets the required standards.

3.1.1.3 Prediction and Analysis

Functionality: This module is used to perform predictions using the trained hybrid model. It includes the visualizations of all the results to the user that include the reports and the time series graph of the electricity demand. The user can use these predictions for load balancing and to analyze the seasonal trends.

Relationship: It is using the previous module that is Model training and updated in which model is trained for the forecasting electricity demand. Then it show visuals to the user.

3.1.1.4 User Interface and Reporting

Functionality: In this module includes all the interface for the users and admins to use the system. It will provide a dashboard to perform predictions to the users and to interact with the system to both users and admins. It is providing a dashboard for visualizing forecasts, generating detailed reports, and interacting with analytical tools to the user. For the admin, it offers data upload capabilities, model monitoring, and management features.

Relationship: This module takes data from the Prediction and Analysis module for forecasting data and to show visuals to user. It also interact with the other modules for the Data Management to upload data and preprocessing the data and Model training and so on.

3.1.1.5 Model Monitoring and Performance

Functionality: The module is used for the track of model performance to test that the model is working correctly or not. It provides the predictions errors and accuracy metrics to update the model on this basis.

Relationship: It connects with Model Training Module to track its performance over time. It gives feedback on accuracy and helping the admin decide when retraining is necessary. Data generated from this module is accessible via the User Interface for review and reporting.

3.1.2 Architecture Diagram

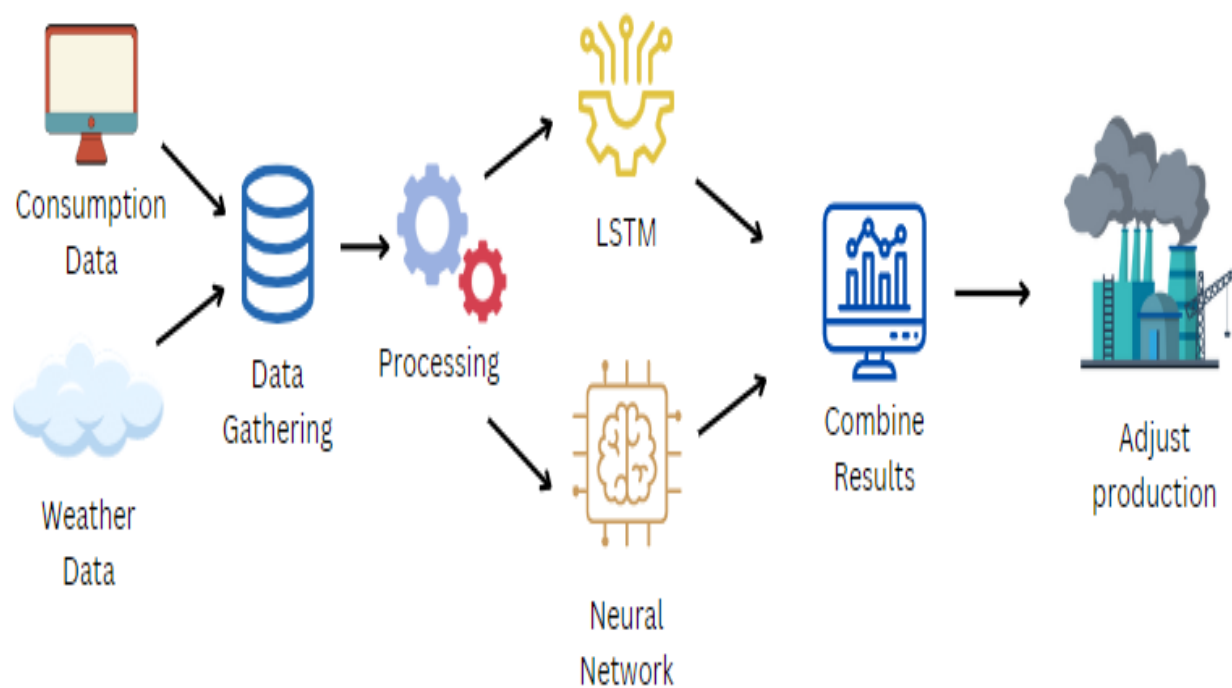


Figure 3.1: Architecture Diagram

3.2 Design Models

3.2.1 Activity Diagrams

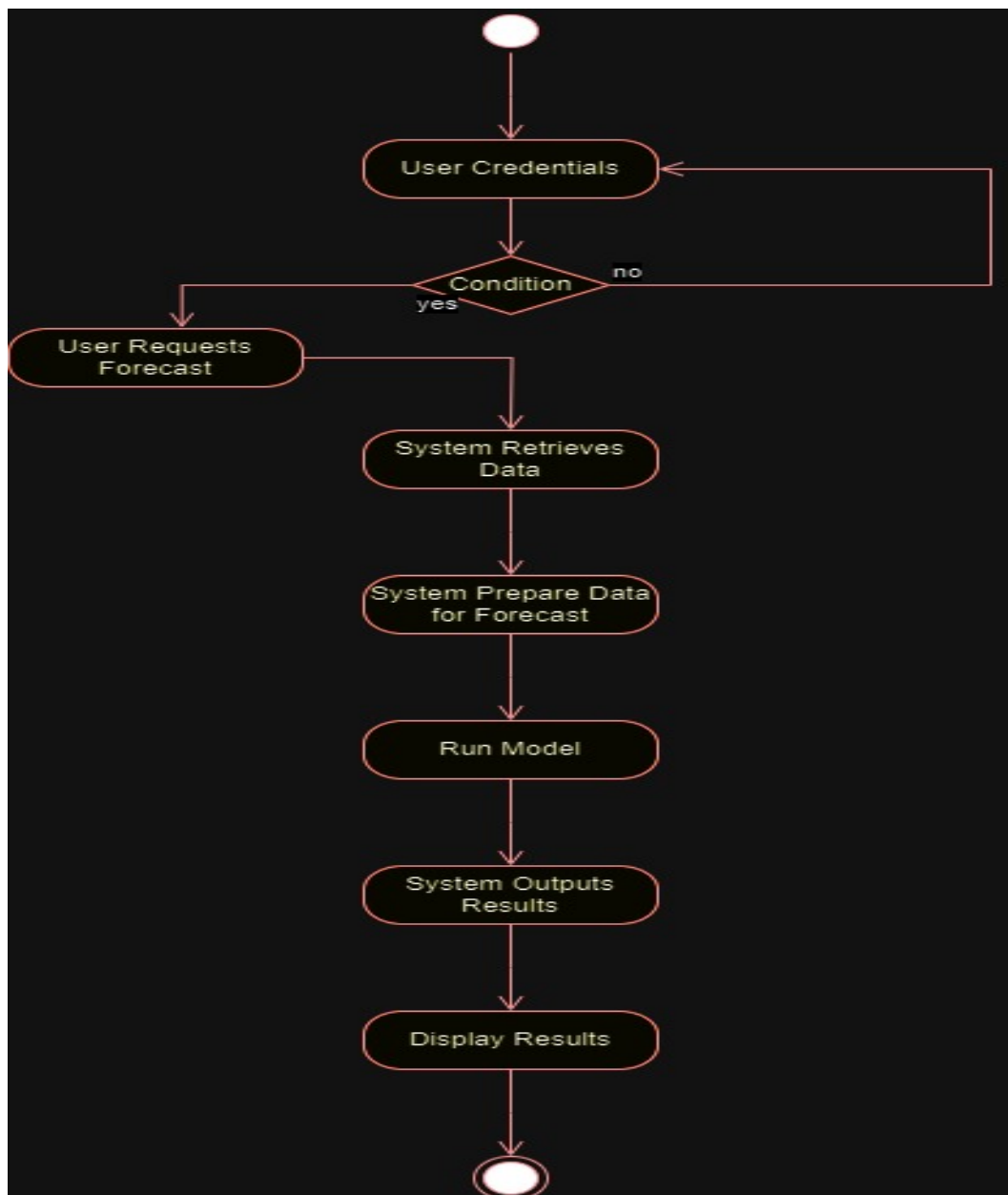


Figure 3.2: Act-1: Forecasting Electricity Demand

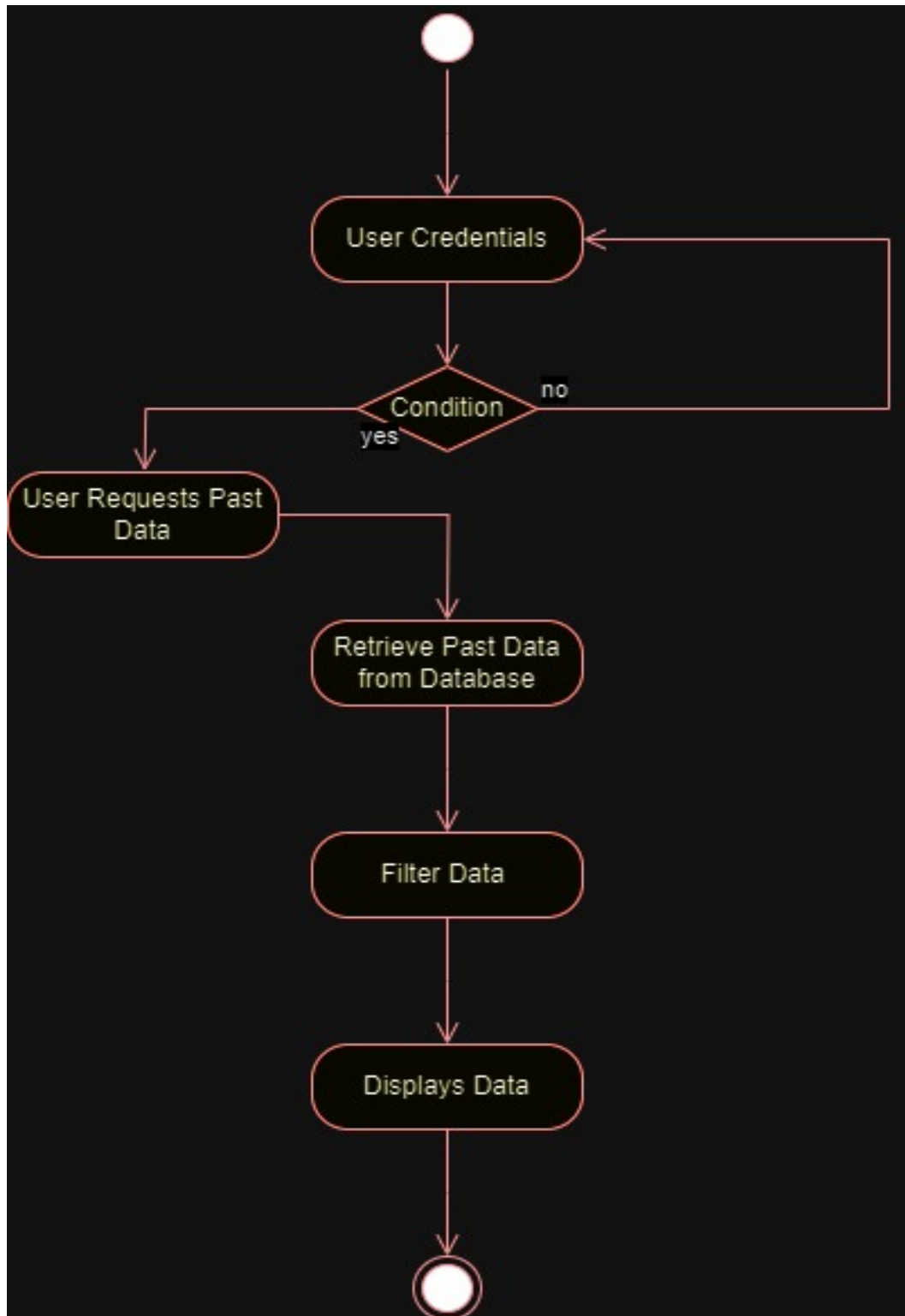


Figure 3.3: Act-2: Historical Predictions

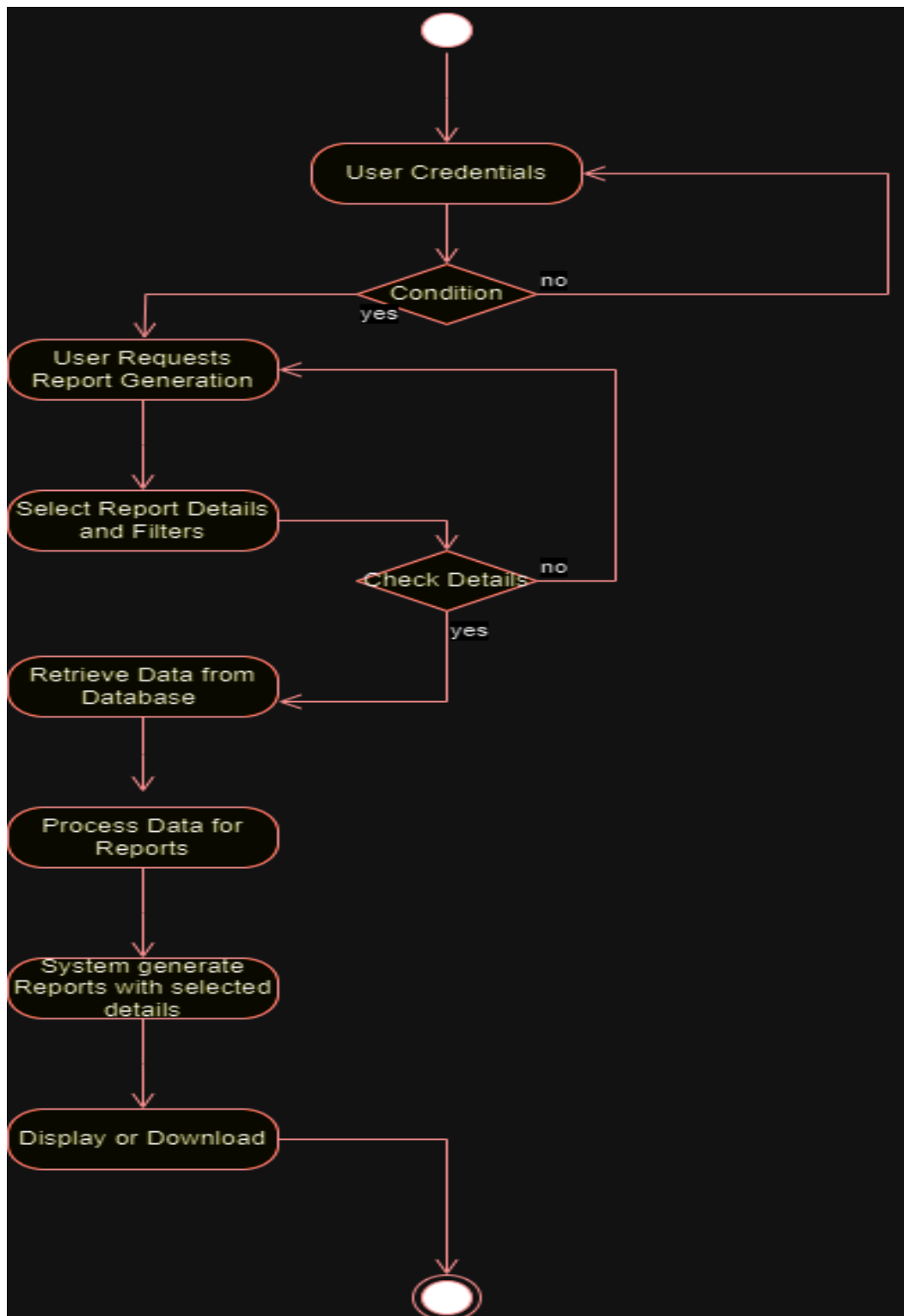


Figure 3.4: Act-3: Generating Detailed Reports

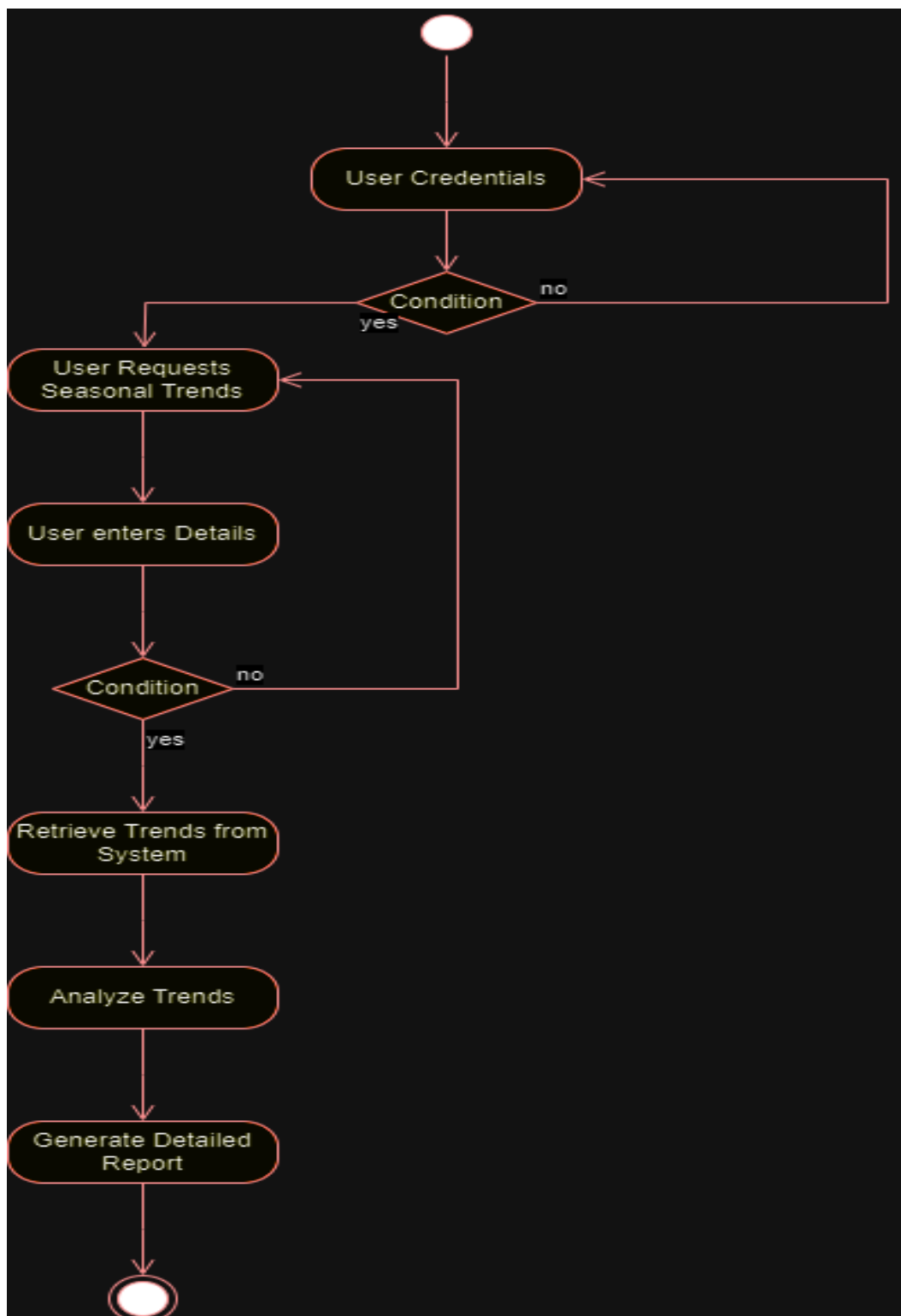


Figure 3.5: Act-4: Analyze Seasonal Trends

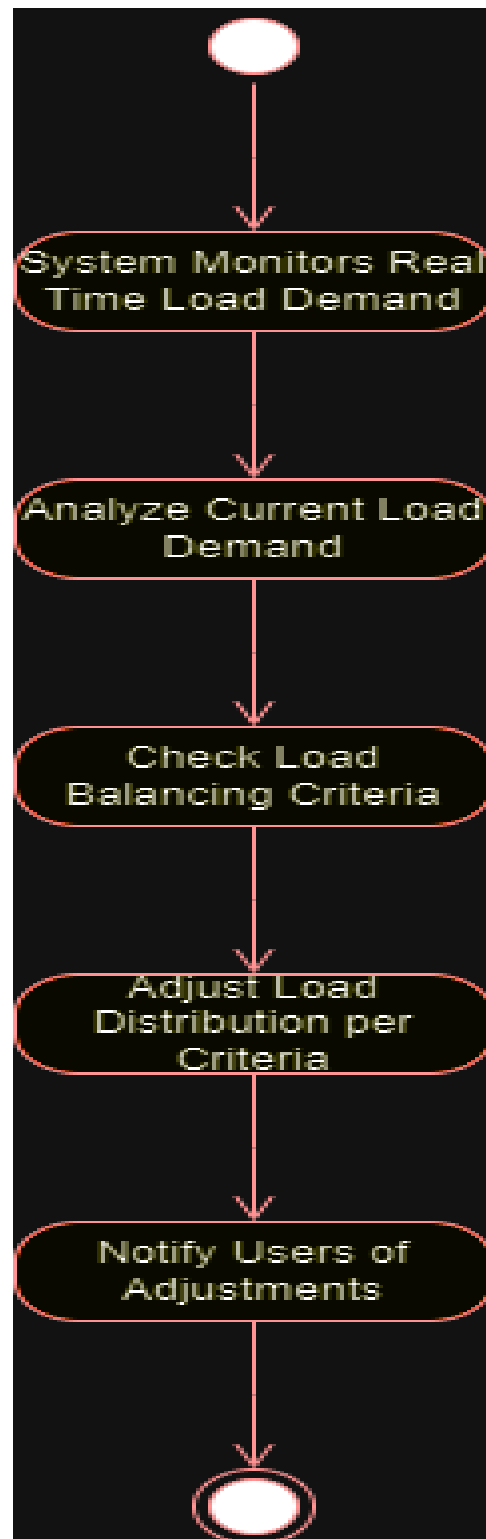


Figure 3.6: Act-5: Realtime Load Balancing

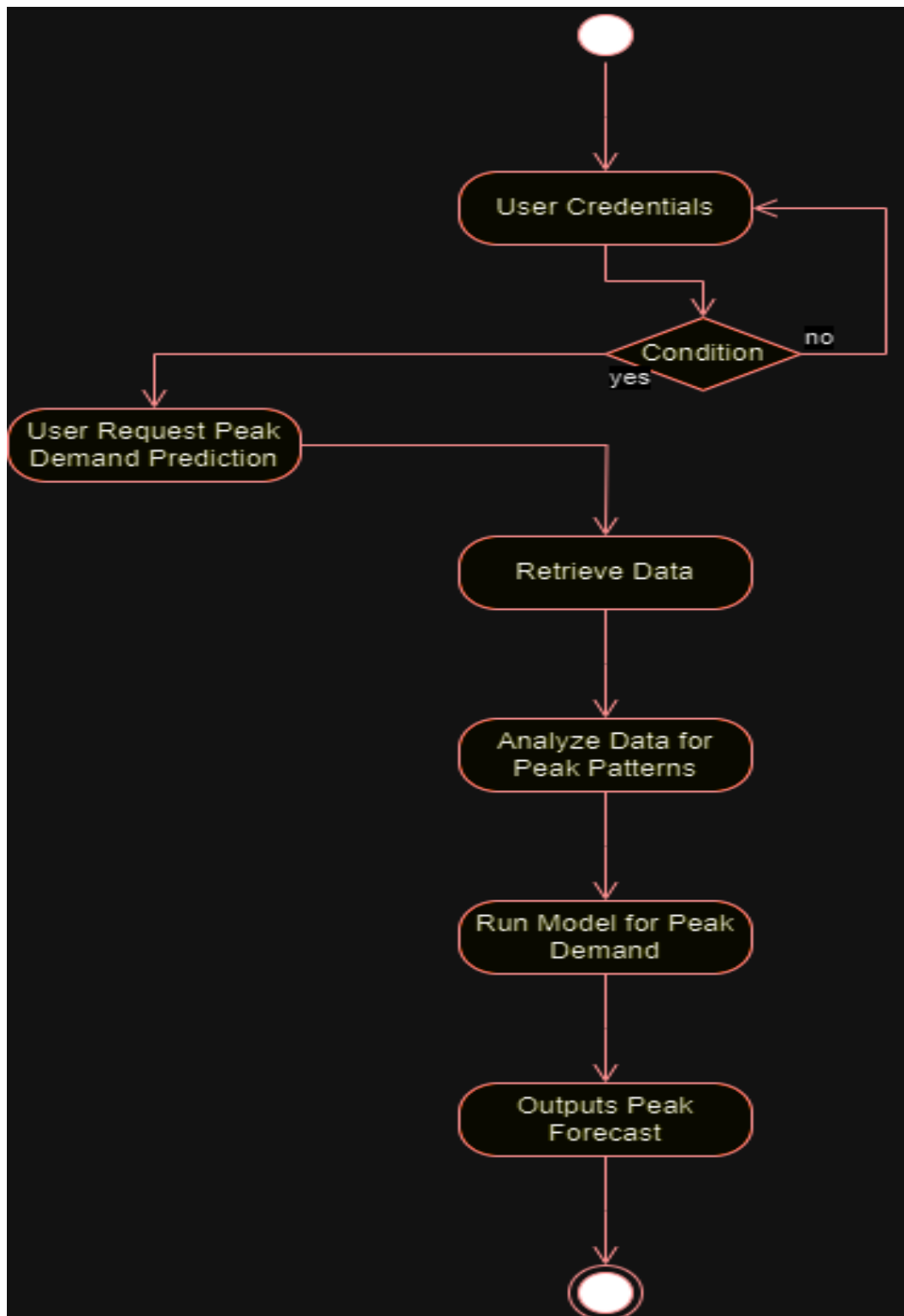


Figure 3.7: Act-6: Peak Demand Prediction

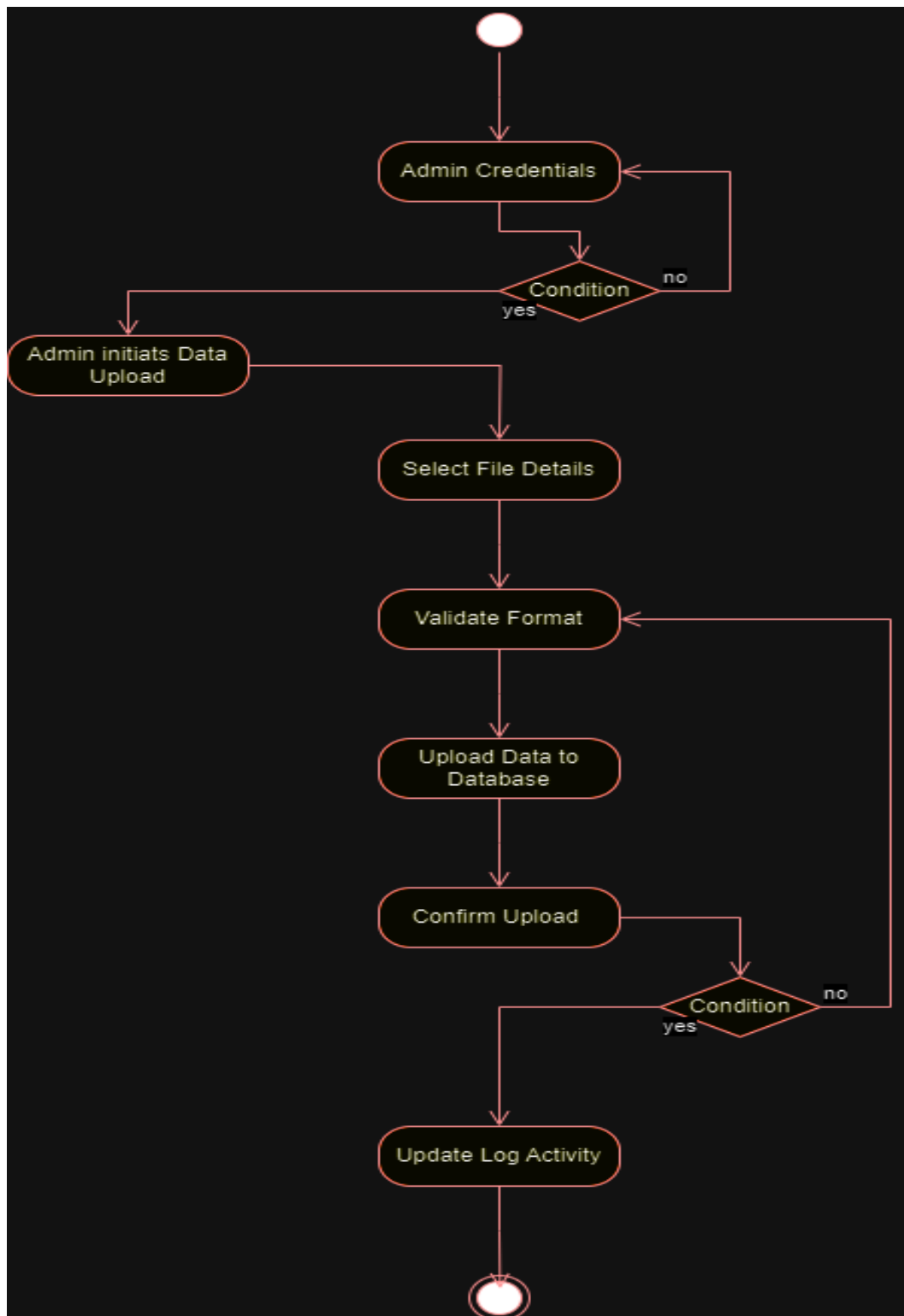


Figure 3.8: Act-7: Upload Historical Data

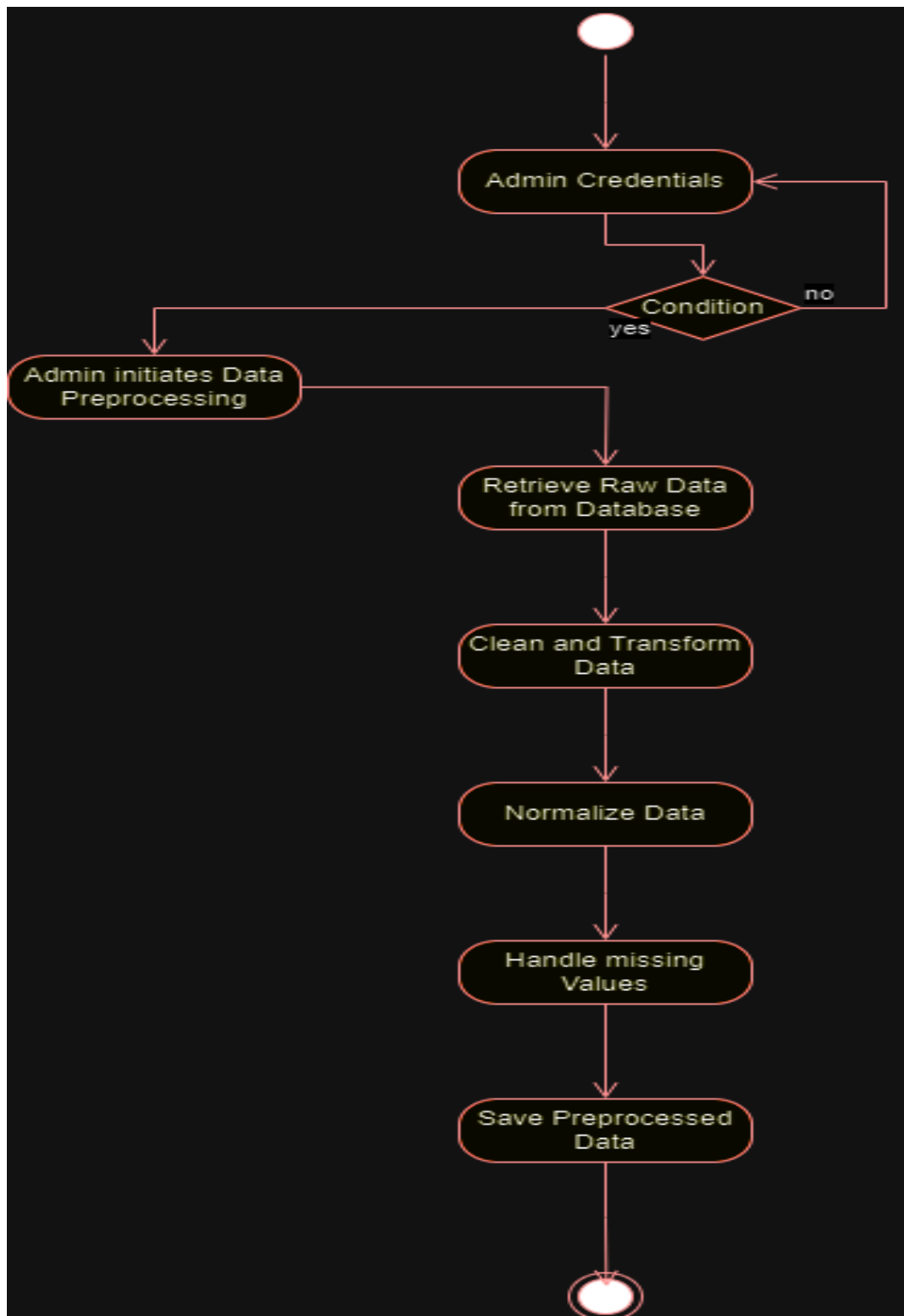


Figure 3.9: Act-8: Data Pre-processing

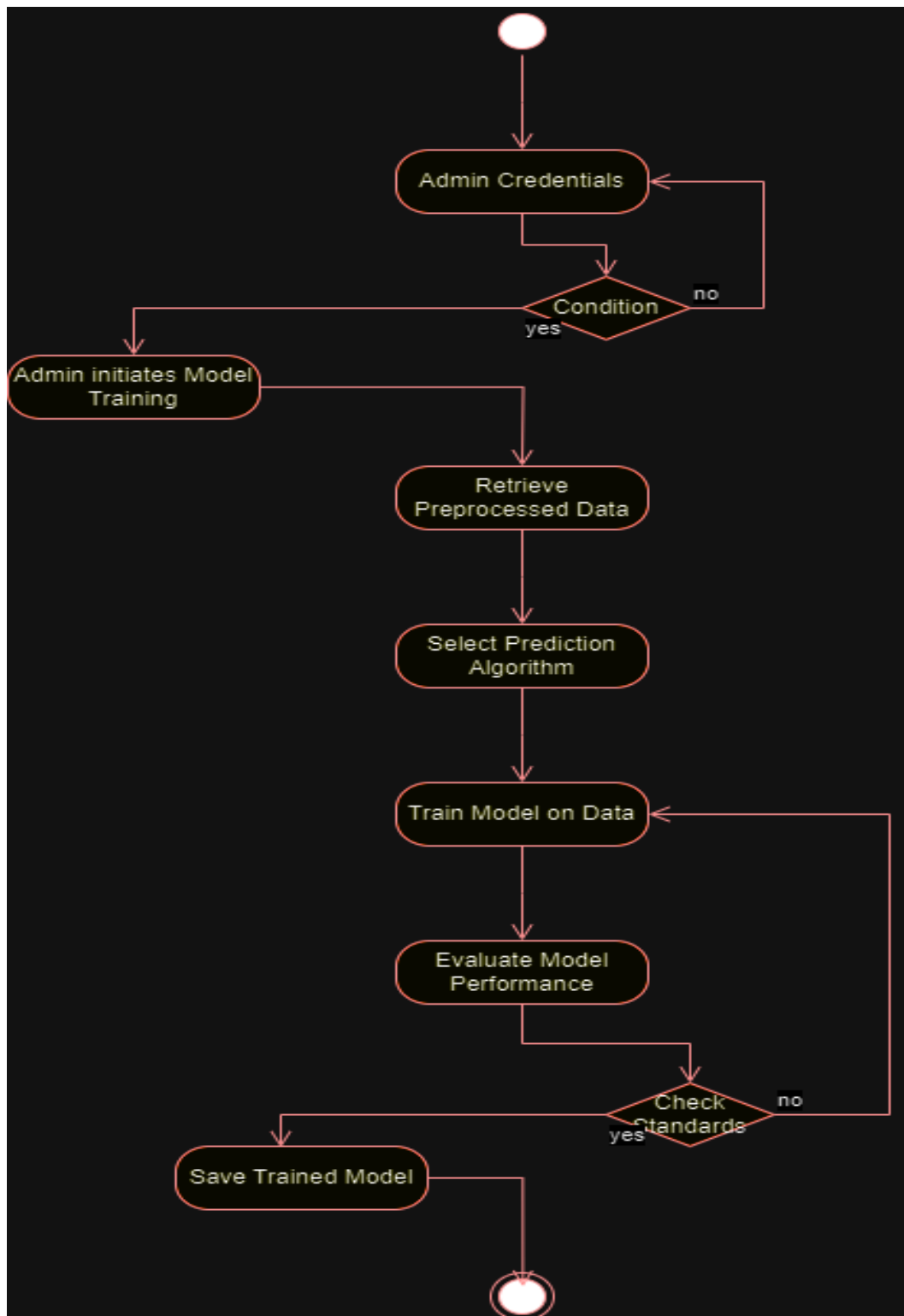


Figure 3.10: Act-9: Train Prediction Model

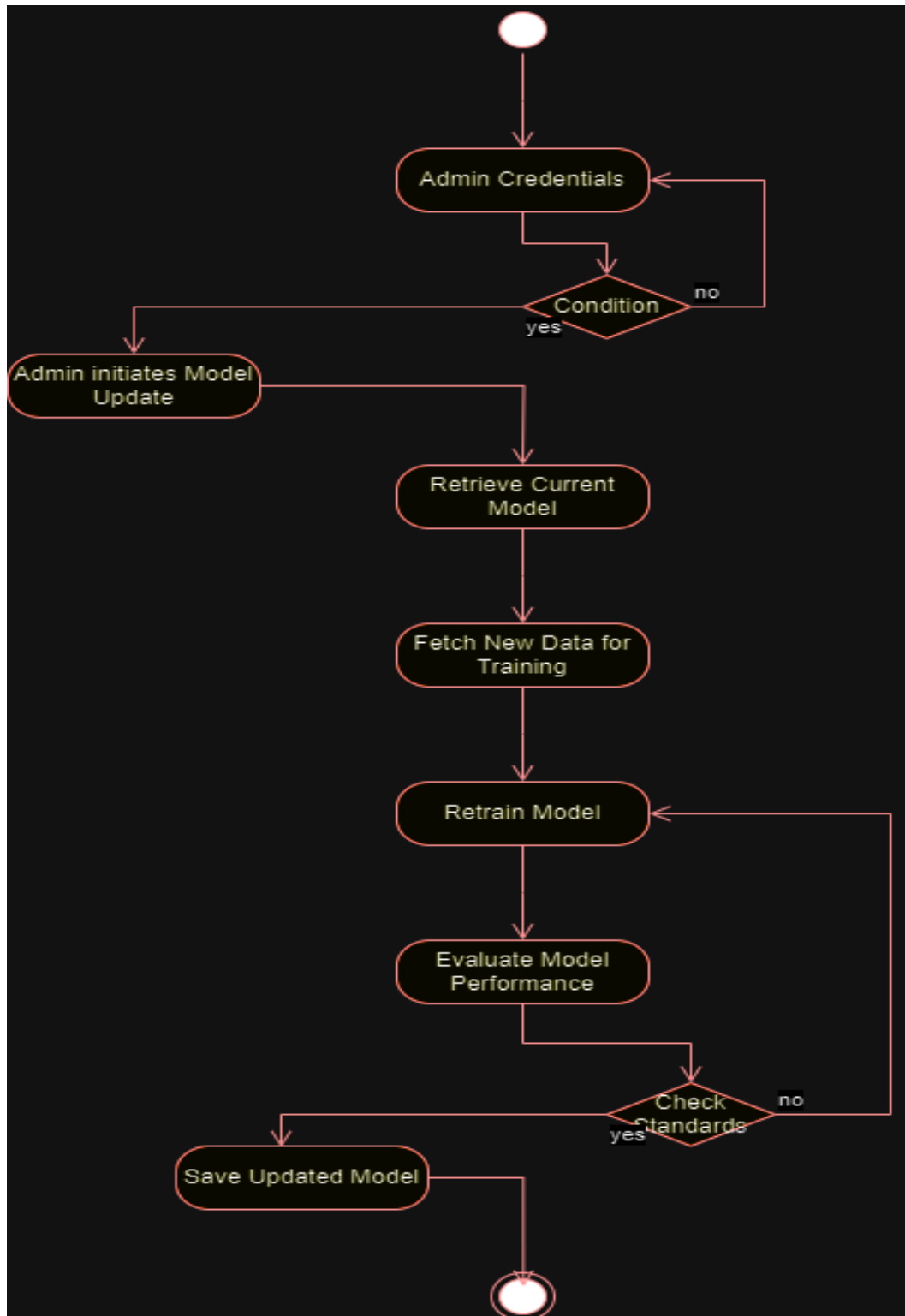


Figure 3.11: Act-10: Update Prediction Model

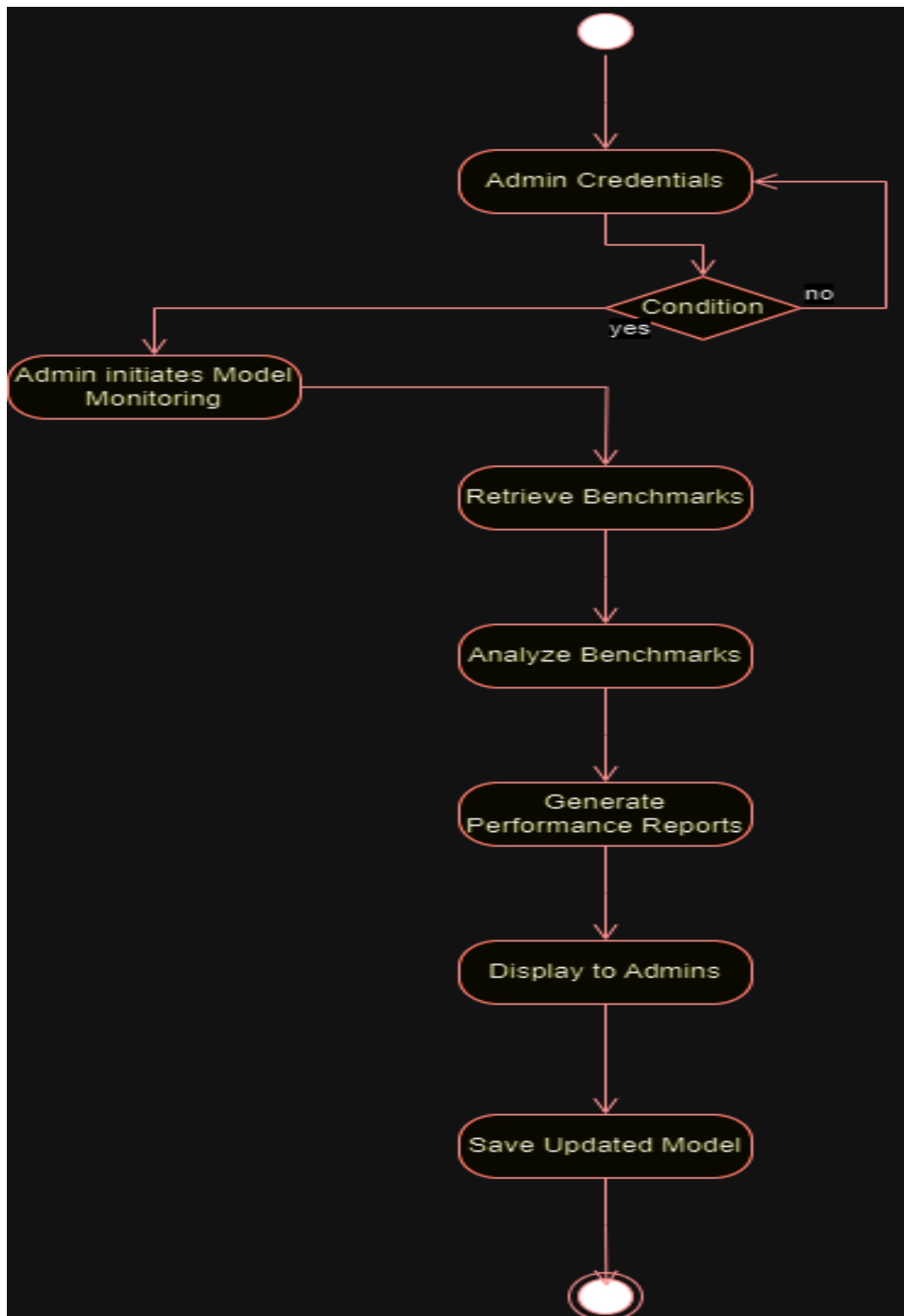


Figure 3.12: Act-11: Monitor Model Performance

3.2.2 Class Diagram

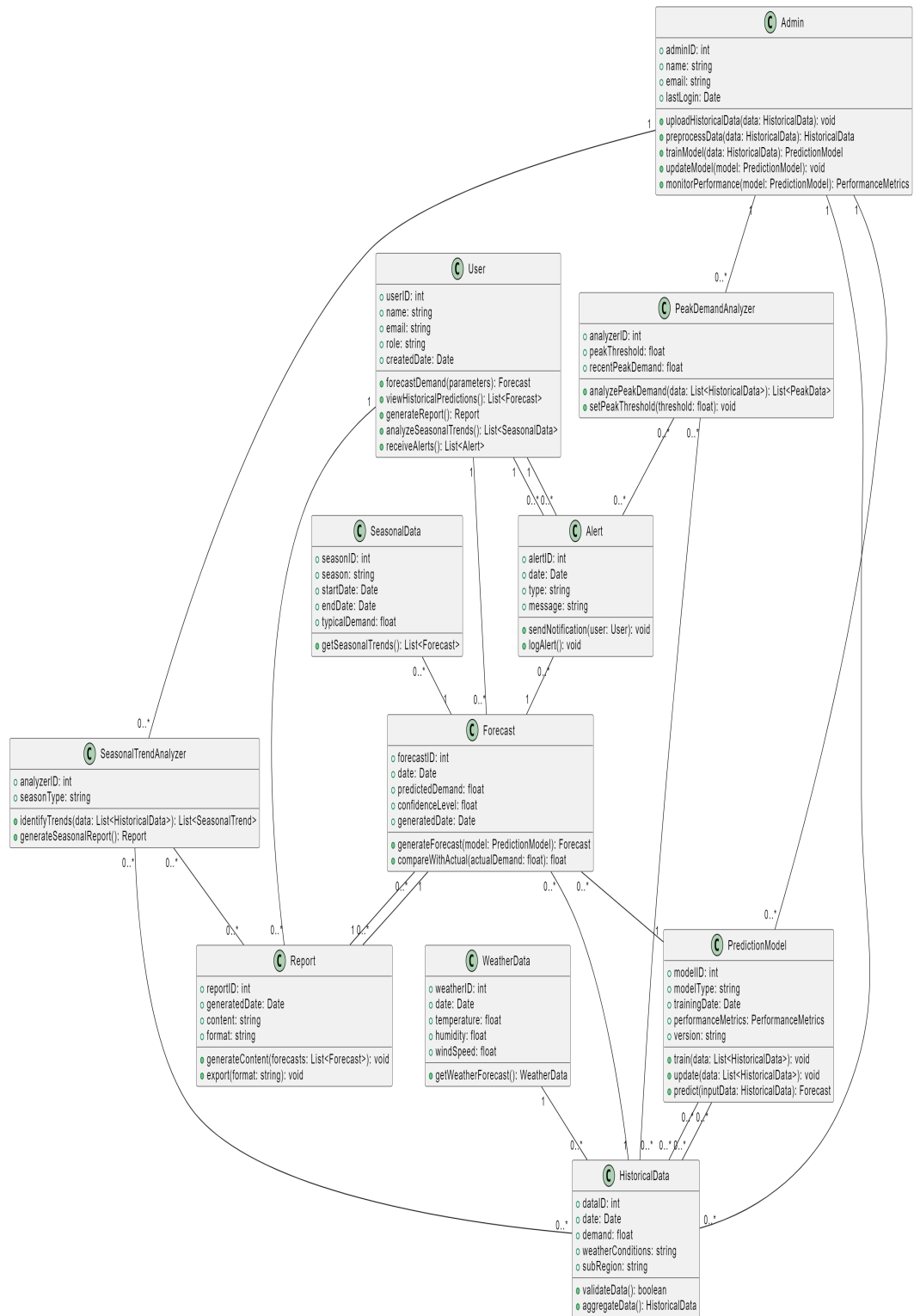


Figure 3.13: Class Diagram

3.2.3 Class-Level Sequence Diagram

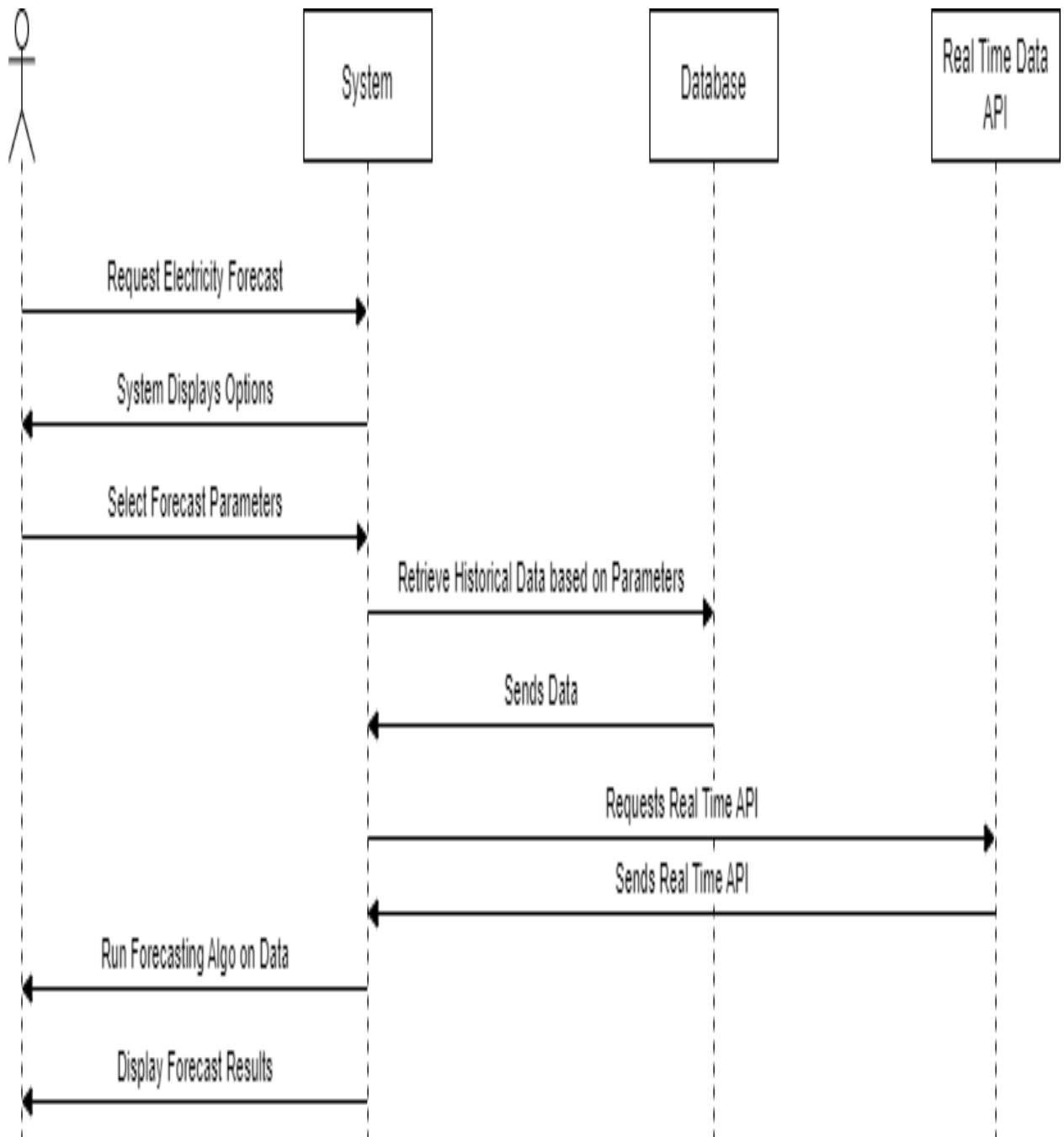


Figure 3.14: ss-1: Forecasting Electricity Demand

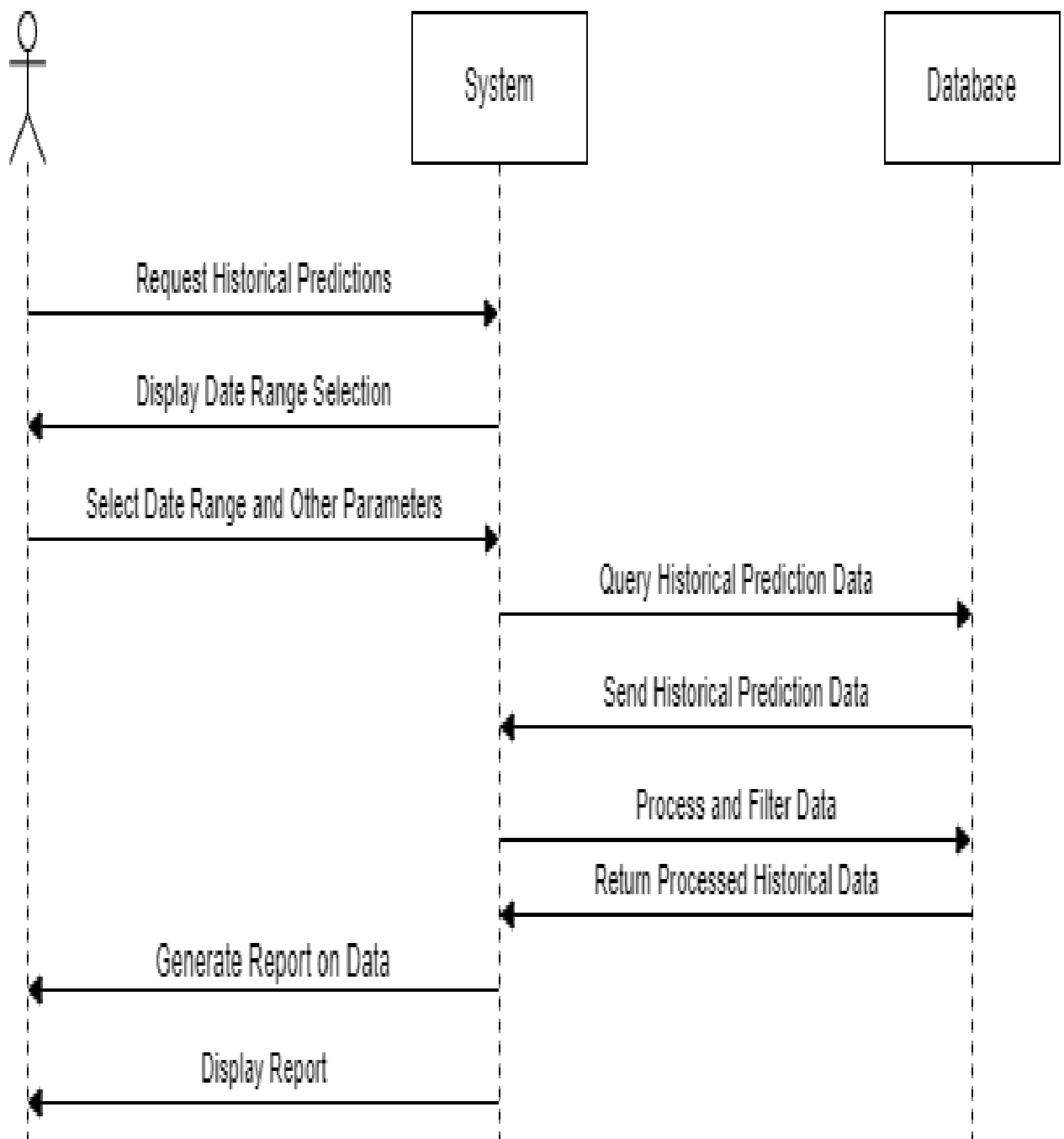


Figure 3.15: ss-2: Historical Prediction

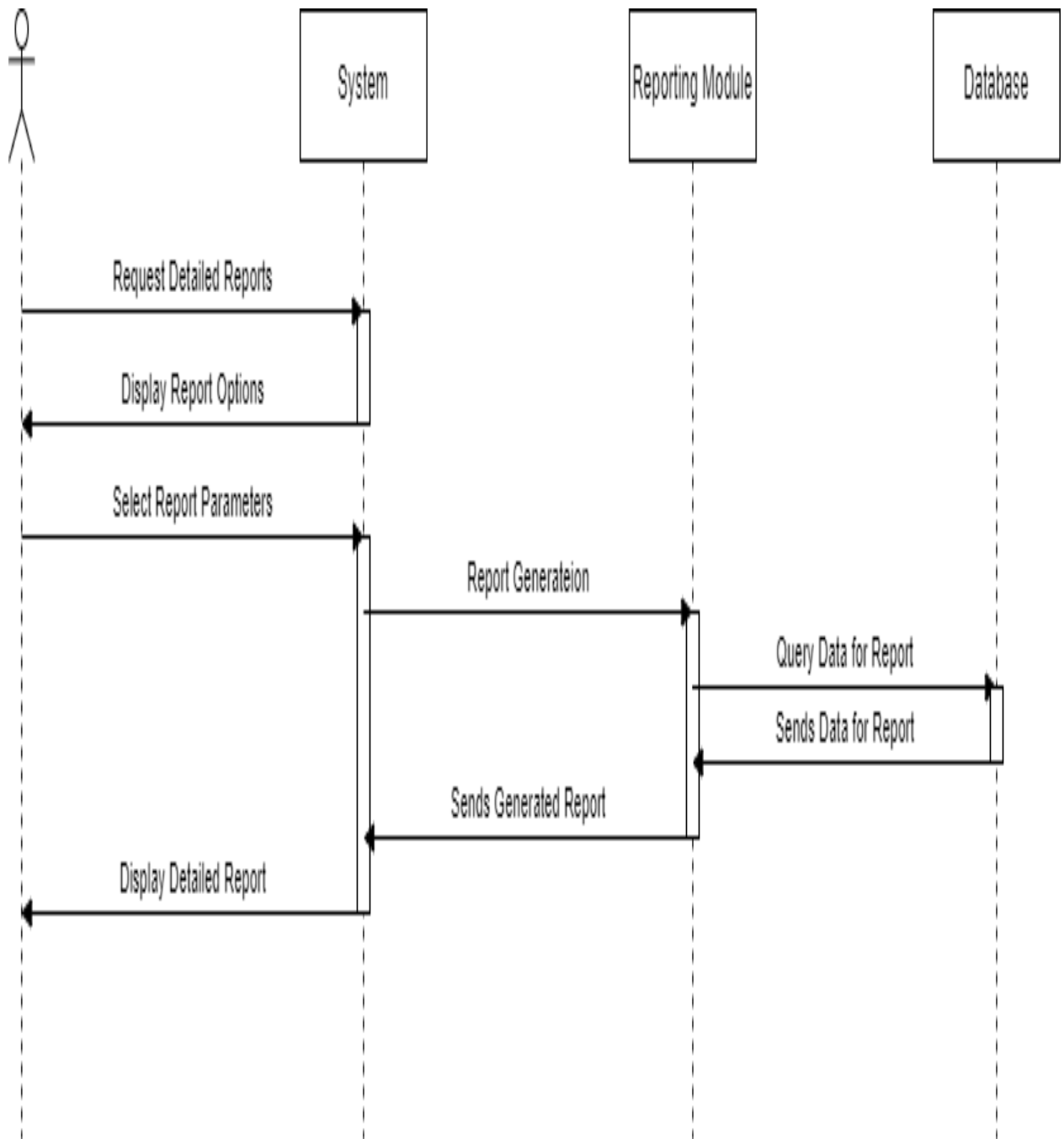


Figure 3.16: ss-3: Generate Detailed Reports

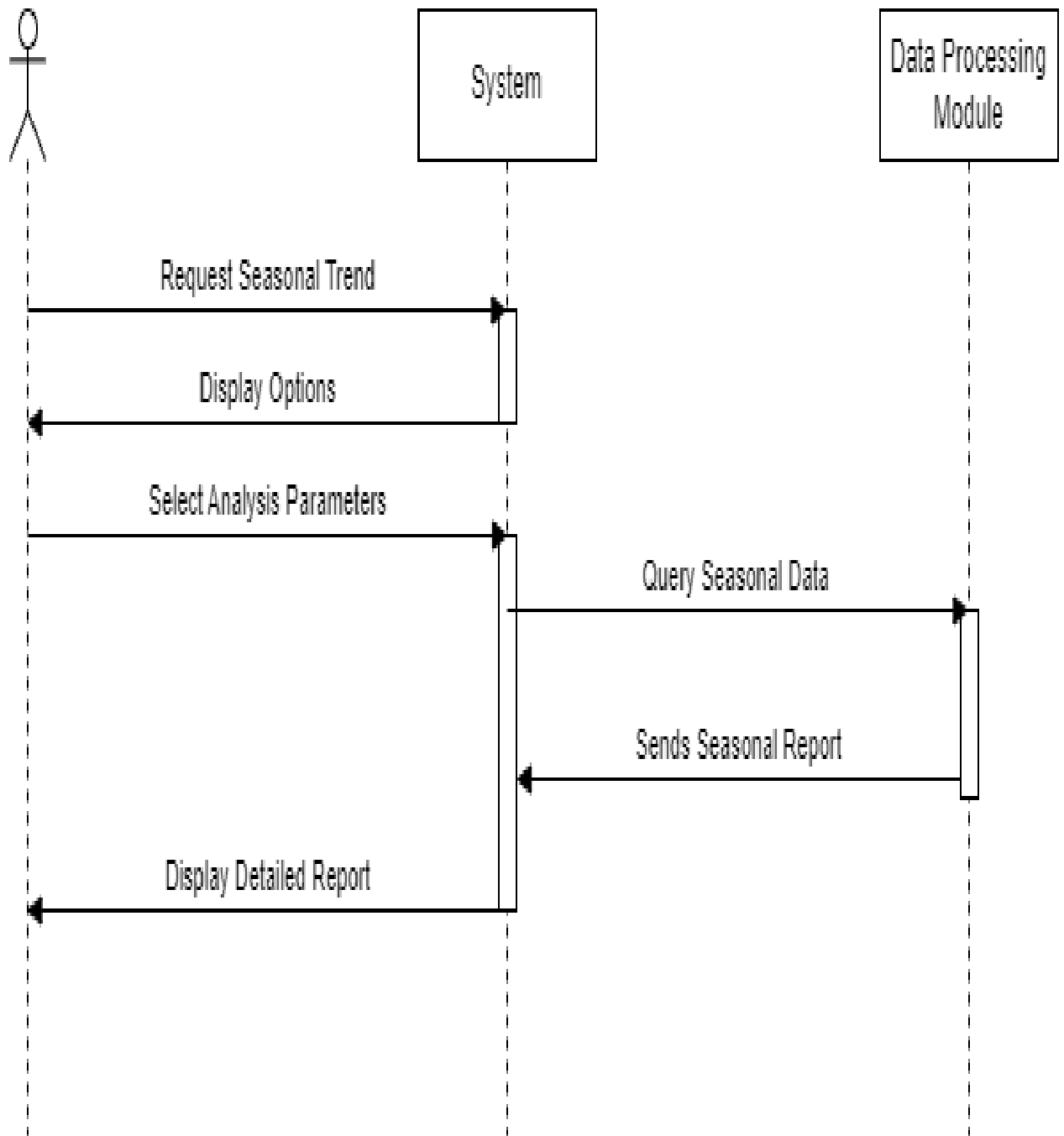


Figure 3.17: ss-4: Analyze Seasonal Trends

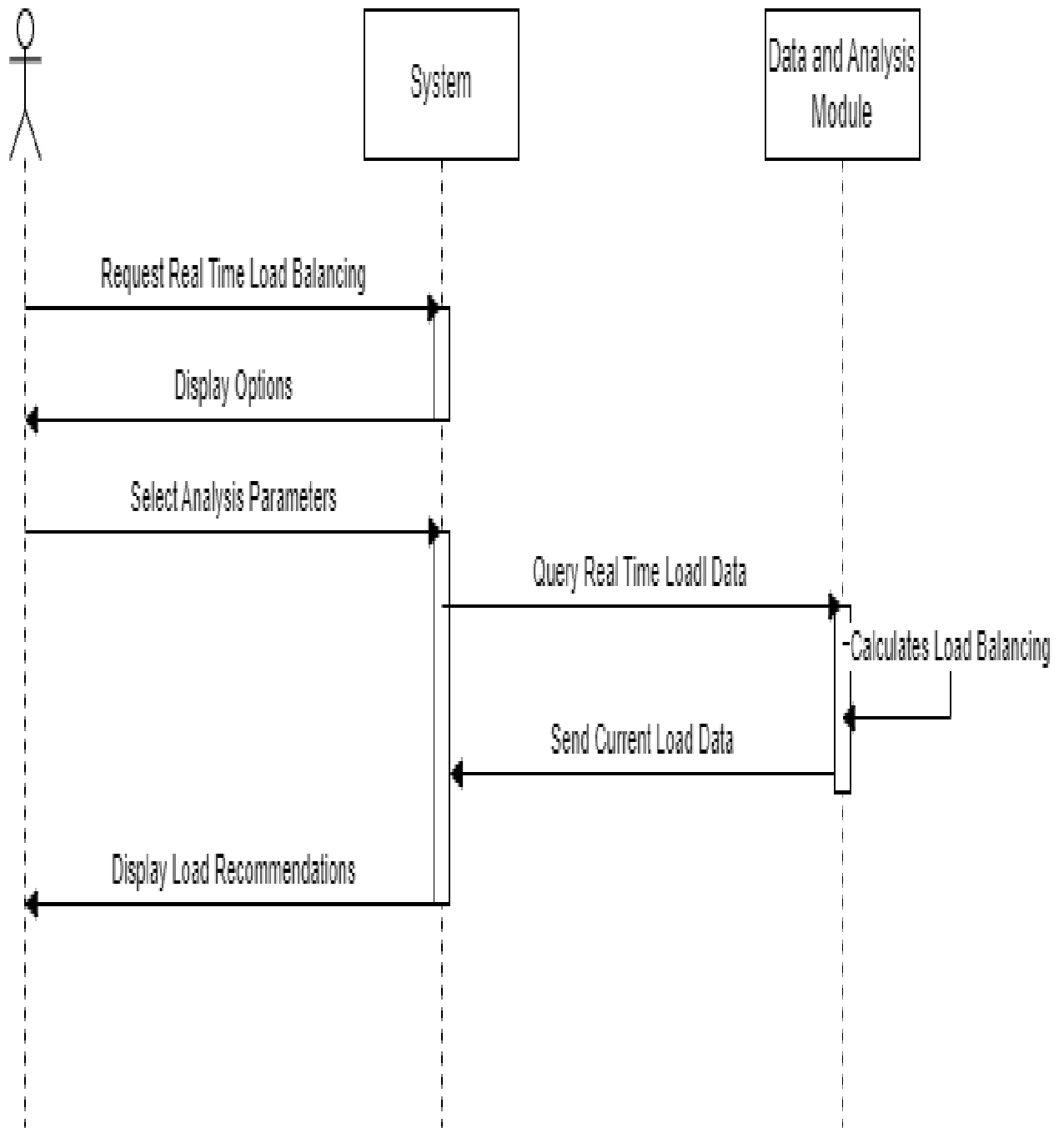


Figure 3.18: ss-5: Realtime Load Balancing

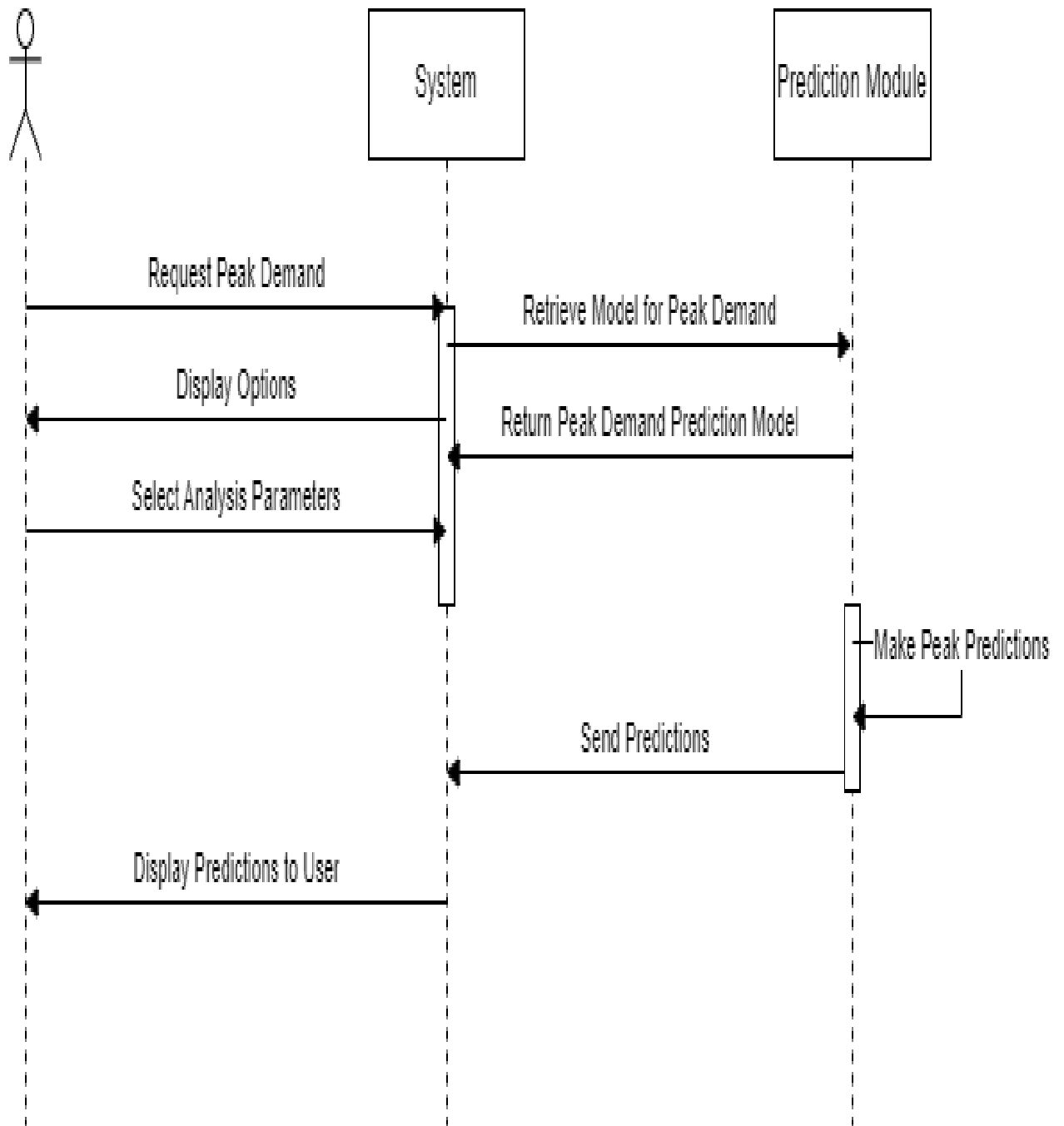


Figure 3.19: ss-6: Peak Demand Prediction

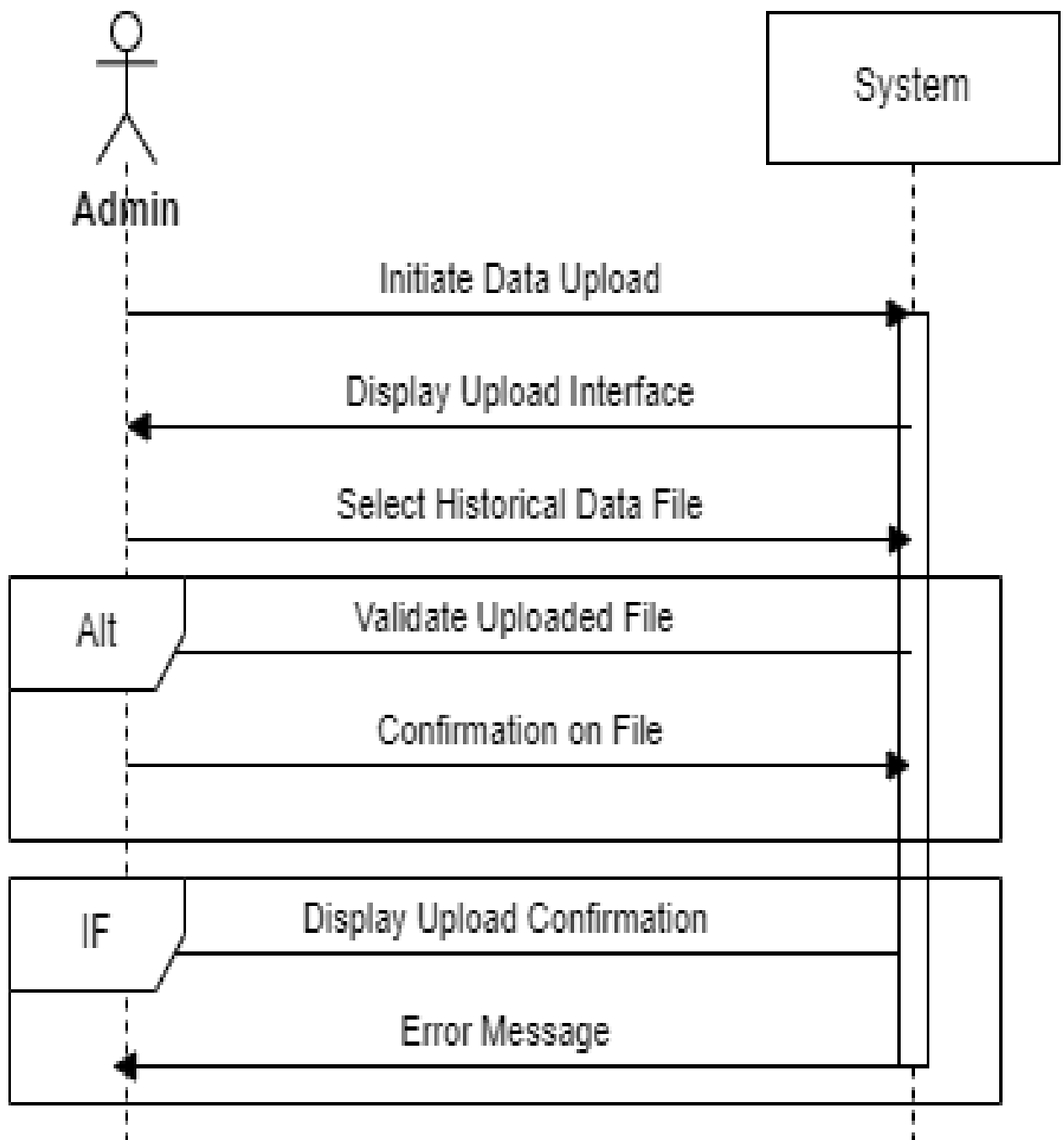


Figure 3.20: ss-7: Upload Historical Data

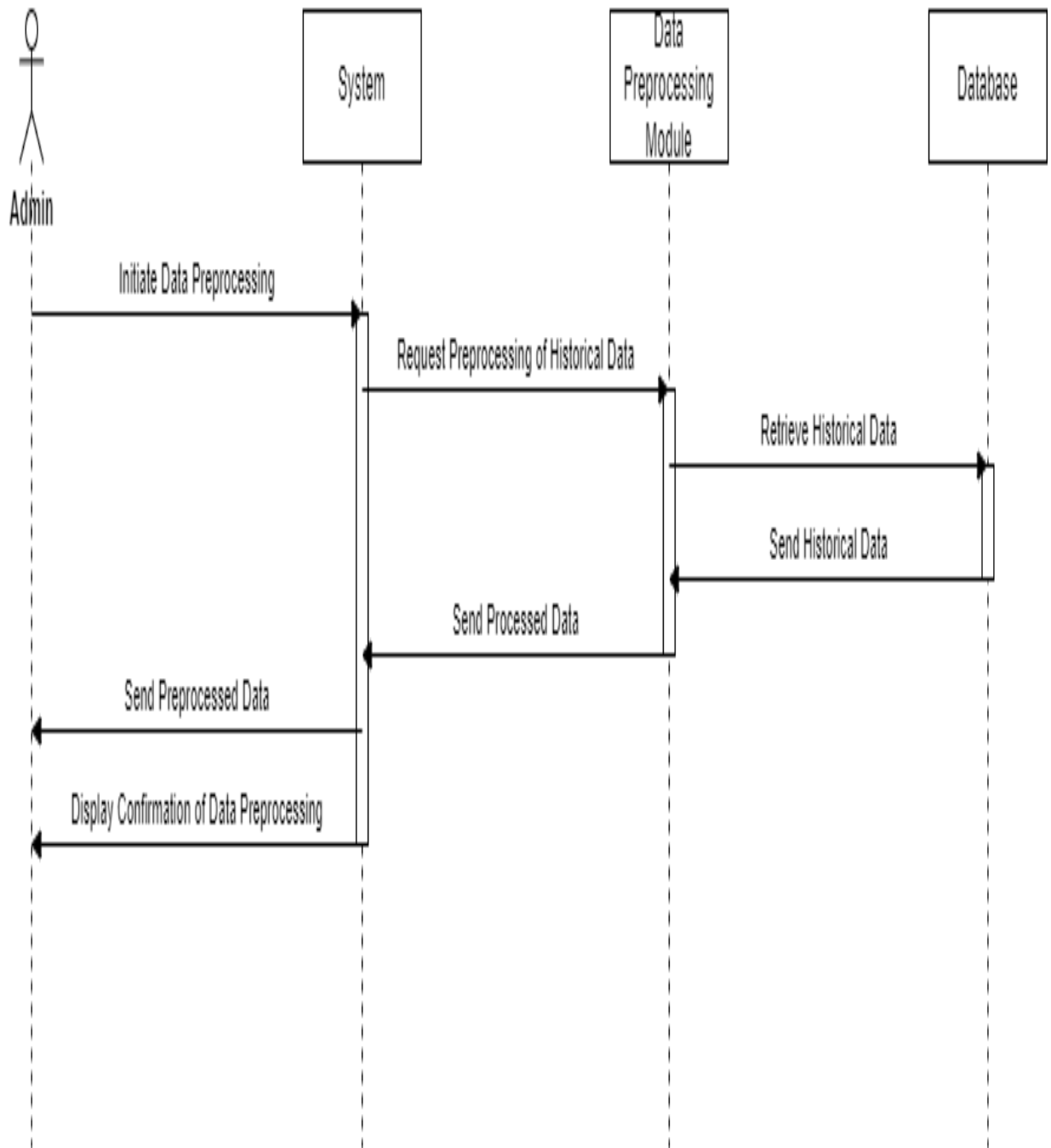


Figure 3.21: ss-8: Data Pre-Processing

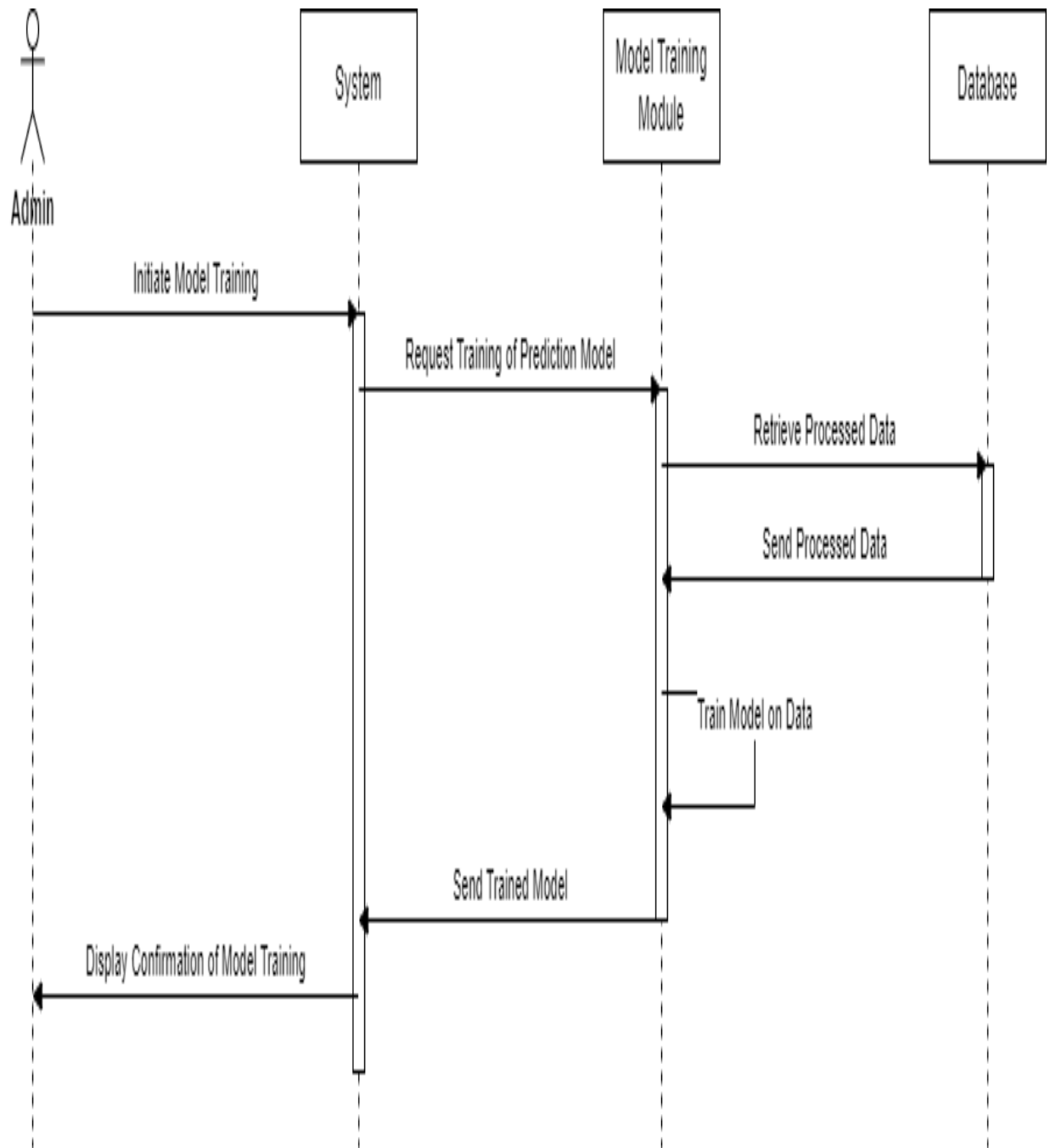


Figure 3.22: ss-9: Train Prediction Model

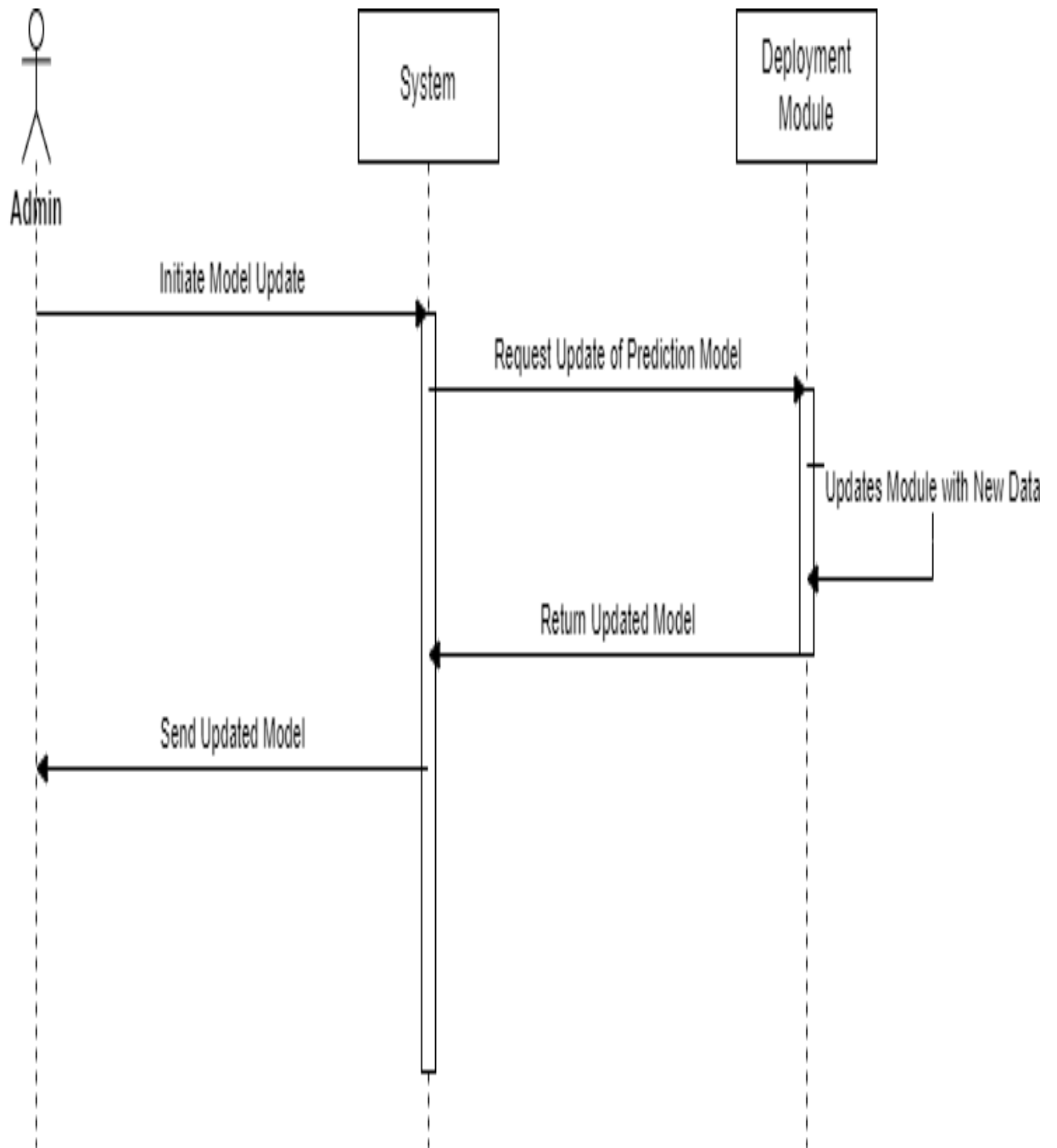


Figure 3.23: ss-10: Update Prediction Model

3.3 Data Design

3.3.1 Data Transformation and Storage

The information ecosystem within 'WattWiseAI' encompasses a variety of data sources, including electricity demand, weather conditions, predictive forecasts, user information, and reports. These data types are structured into specific formats to optimize how they are stored, processed, and retrieved, supporting the system's forecasting and reporting capabilities.

3.3.2 Key Data Entities and Their Storage

3.3.2.1 Users

- **Data Structure:** User details are organized in a relational database, where each user has fields like UserID, Name, Email, AccessLevel, and Preferences.
- **Storage and Processing:** User data is housed in a centralized database (e.g., PostgreSQL) to facilitate authentication and role-based access control. Additionally, activity logs are maintained to customize user experiences and generate personalized reports.

3.3.2.2 Electricity Demand Data

- **Data Structure:** This is stored in a time-series format with attributes such as Date, Hour, BalancingAuthority, SubRegion, and DemandMW.
- **Storage and Processing:** A time-series database (e.g., InfluxDB or TimescaleDB) is used to store this data, making it efficient for time-based retrieval and analysis. Historical demand patterns inform the system's forecasting models to generate future predictions.

3.3.2.3 Weather Data

- **Data Structure:** Weather data records include factors like Temperature, Humidity, WindSpeed, and SeasonIndicator, each tied to a specific time and location.
- **Storage and Processing:** Stored in a relational database with indexed fields for fast querying, this data is regularly integrated with electricity demand data for preprocessing, feeding into the forecasting model for training and predictions.

3.3.2.4 Forecasting Model

- **Data Structure:** Model data consists of parameters and configurations, stored under fields like ModelID, ModelType (e.g., ANN, EDLSTM, ICEEMDAN), Parameters, and TrainingDate.
- **Storage and Processing:** These configurations and model histories are stored in a NoSQL database (e.g., MongoDB), allowing for flexible storage that can easily adapt to varied model parameters and enable fast access for retraining or updates.

3.3.2.5 Predictions

- **Data Structure:** Prediction records include PredictionID, Date, PredictedDemandMW, SubRegion, and AccuracyMetrics.
- **Storage and Processing:** Stored in a time-series database, predictions are aligned with the structure of historical data and are used in the interface for trend analysis and demand management purposes.

3.3.2.6 Reports

- **Data Structure:** Reports are composed of fields like ReportID, UserID, ReportType, GeneratedDate, and Content.
- **Storage and Processing:** Generated on demand, these reports are stored in a document-oriented database (e.g., MongoDB) for quick and easy retrieval. Reports support various types, including historical trends, seasonal analyses, and prediction evaluations.

3.3.2.7 Seasonal Trends

- **Data Structure:** Seasonal trends are captured in fields like TrendID, SeasonType, AverageDemand, and YearlyChange.
- **Storage and Processing:** Trends derived from historical data are stored in a relational database, enabling the analysis of seasonal patterns. This data is periodically updated to help improve the accuracy of the forecasting models.

3.3.3 Database Storage and Solutions

- **Relational Database:** PostgreSQL is used to store structured data, such as user profiles and reports, ensuring data consistency and allowing efficient SQL queries.
- **Time-Series Database:** InfluxDB or TimescaleDB handles time-series data like electricity demand and predictions, ideal for managing chronological datasets.
- **NoSQL Database:** MongoDB is used for storing non-relational data, such as model configurations and parameters, offering dynamic and flexible storage.
- **Document Storage:** Amazon S3 or Google Cloud Storage is utilized for storing large documents, such as detailed reports and seasonal trend analyses, in a scalable and durable way.

3.3.4 ERD

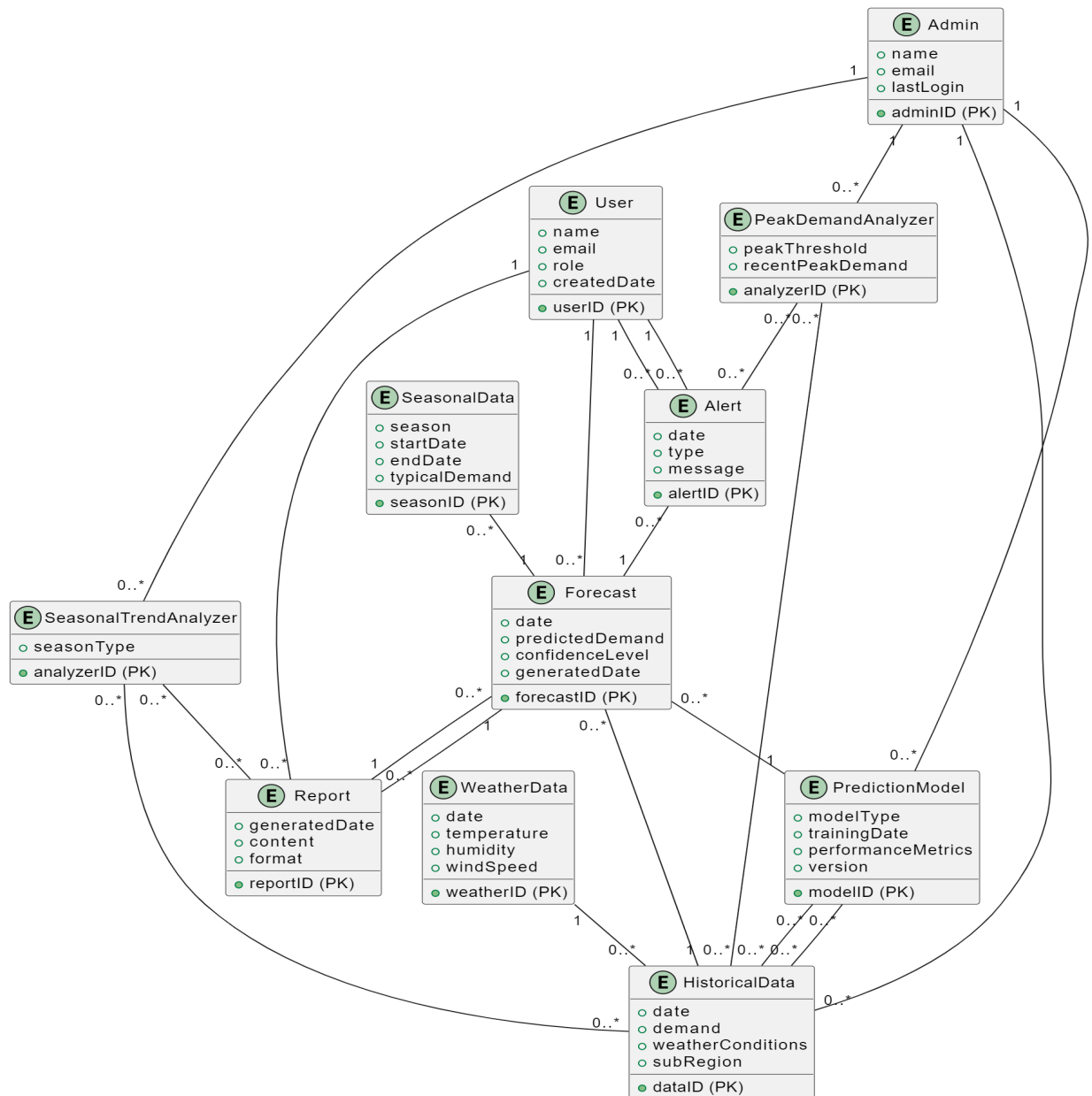


Figure 3.24: ERD

Chapter 4

Conclusions and Future Work

The WattWiseAI project indicates the sophistication of data analysis and forecasting techniques that serve as an important basis for the better management of electricity demand. In developing an efficient and accurate daily electrical demand prediction system, WattWiseAI can show the inadequacies that exist in balancing energy supply and consumption needs at utility companies. It combines historical consumption patterns with external factors, such as weather and seasonality, to obtain actionable insights that will help reduce forecasting errors, optimize grid stability, and provide more precise resource allocation. If that is not enough, WattWiseAI provides user and admin functionality for frictionless forecasting, reporting, and real-time monitoring so stakeholders make truly informed decisions. This positions WattWiseAI for both promoting sustainability in energy management and reducing operational cost, hence a step towards resilience and adaptability in power grid solutions for the future.

Bibliography

1. <https://www.sciencedirect.com/science/article/pii/S0196890423010531>
2. <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2024.1442502/full>
3. <https://econjournals.com/index.php/ijeep/article/view/16506>
4. https://www.researchgate.net/publication/271364922_Electricity_Consumption_Forecasting_in_Thailand_Using_an_Artificial_Neural_Network_and_Multiple_Linear_Regression